

Data-Driven Distributionally Robust CVaR Portfolio Optimization under Regime-Switching Ambiguity Set

Chi Seng Pun

School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore, cspun@ntu.edu.sg,
<https://personal.ntu.edu.sg/cspun/>

Tianyu Wang

Department of Industrial Engineering and Operations Research, Columbia University, United States, tw2837@columbia.edu

Zhenzhen Yan*

School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore, yanzz@ntu.edu.sg,
<https://sites.google.com/site/yanzhenzhen888/>

* indicates a corresponding author

Problem definition: Non-stationarity of the random environment is a critical yet challenging concern in decision-making under uncertainty. We illustrate the challenge from the non-stationarity and the solution framework using the portfolio selection problem, a typical decision problem in a time-varying financial market. **Methodology/Results:** This paper models the non-stationarity by a regime-switching ambiguity set. In particular, we incorporate the time-varying feature of the stochastic environment into the traditional Wasserstein ambiguity set to build our regime-switching ambiguity set. This modeling framework has strong financial interpretations since the financial market exposes to different economic cycles. We show that the proposed distributionally optimization framework is computationally tractable. We further provide a general data-driven portfolio allocation framework based on a covariate-based estimation and a hidden Markov model. We prove that the approach can include the underlying distribution with a high probability when the sample size is larger than a quantitative bound, from which we further analyze the quality of the obtained portfolio. Extensive empirical studies are conducted to show that the proposed portfolio consistently outperforms the equally-weighted portfolio (the $1/N$ strategy) and other benchmarks across both time and datasets. In particular, we show that the proposed portfolio exhibits a prompt response to the regime change in the 2008 financial crisis by reallocating the wealth into appropriate asset classes on account of the time-varying feature of our proposed model. **Managerial implications:** The proposed framework helps decision-makers hedge against time-varying uncertainties. Specifically, applying the proposed framework to portfolio selection problems helps investors respond promptly to the regime change in financial markets and adjust their portfolio allocation accordingly.

Key words: Time-varying uncertainty, Regime-switching ambiguity, Hidden Markov model, Portfolio selection

1. Introduction

A typical challenge in decision-making with uncertainty is the changing random environment. For instance, in e-commerce, demands can be highly non-stationary, changing with seasons or even the time of day. In supply chains, the high volatility of disruption results in a significant non-stationarity of the market during a disruption. The time-varying stochastic environment is especially relevant in the application of portfolio optimization since economic conditions change from time to time. Bridgewater (2012) revealed that different asset classes do well in certain regimes and poorly in others. Before that, they created the All-Weather strategy in 1996, which allocated portfolios via the risk of each asset class, known as risk parity. Various funds, such as the Canadian Pension Fund, gradually recognized the benefits of asset allocation via risky parity and adopted this strategy since then. In this paper, we focus on the portfolio selection problem to illustrate the challenge of the time-varying uncertainties and our solution framework.

Modern portfolio theory, pioneered by Markowitz (1952), is central to finance and is the building block of many economic models. Portfolio optimization is the process of looking for the optimal balance between the reward and risk of the portfolio, which is traditionally measured by the mean and the variance of the portfolio's return. The inferior out-of-sample performance of the mean-variance portfolio (see Michaud 1989, DeMiguel et al. 2009) attracts many proposals of remedies to the estimation error, such as the robust control approach (see Hansen et al. 2006, Fouque et al. 2016, Pun 2021) and the use of alternative risk measures. Value-at-risk (VaR) and conditional value-at-risk (CVaR) are two popular quantile-based risk measures in the finance industry, and they are widely studied in many other fields such as engineering and economics. CVaR is, however, considered a better risk measure than VaR for portfolio optimization due to its coherent property. In addition, CVaR can be expressed as a minimization formula while preserving convexity (see Rockafellar and Uryasev 2002). However, the inaccurate estimations of the returns' distribution would produce a poor out-of-sample performance of the constructed portfolio (Litterman 2003).

To overcome the risk of inaccurate estimations due to limited historical data, distributionally robust optimization (DRO) approaches are deployed. DRO allows the decision maker to optimize against the worst-case distribution over a set of distributions with only

partial statistical information inferred from the available data. We name the set of distributions an ambiguity set. Specifically, a statistical-distance-based ambiguity set has been widely studied recently. This type of ambiguity set is constructed to contain all the distributions centering around a nominal distribution (center) within a certain distance threshold, where the distance can be measured by various metrics such as ϕ -divergence and Wasserstein metric that is of interest in this paper. In the application of portfolio allocation, this center is often chosen as the empirical distribution (see Pflug et al. 2012, Blanchet et al. 2022), which converges to the true distribution as the sample goes to infinity if the returns of assets are assumed to be i.i.d. across time. However, this assumption is often violated in real applications due to the nonstationary nature of the decision-making environment.

Various data-driven optimization approaches are proposed to cope with the nonstationary nature of the decision-making environment. One stream of literature considers a set of distributions across time to capture the possible evolution of uncertainties. Among them, Markov-chain-based models are favored in capturing the evolution of uncertainties (see Zhou et al. 2020, Li et al. 2021). Our paper models the time-varying uncertainties using a regime-switching ambiguity set. In particular, the distribution of the uncertain return depends on different regimes, which correspond to different market conditions. Regimes evolve over time following a Markov chain. Under each regime, a Wasserstein-distance-based ambiguity set is constructed to capture a family of distributions that are close to the empirical estimates of returns. We propose two approaches to estimate the ambiguity set: (1) pre-determining the regimes using covariates (such as macroeconomic variables) and using frequency to estimate the transition probabilities; (2) applying a hidden Markov model (HMM) to automatically calibrate regimes and transition probabilities from data. To the best of our knowledge, this paper is the first to derive a general data-driven framework for regime-switching models in a DRO setting.

We summarize the contributions of this paper as follows. First, we generalize the white paper of Bridgewater (2012) to divide the financial market into four different economic conditions and then estimate regimes and the corresponding transition probabilities matrix using two methods: (1) pre-determining regimes from covariates and estimating transition probabilities using the frequency of occurrence from one regime to another; (2) HMM. Subsequently, we incorporate the regime-switching nature of asset returns into the distributionally robust portfolio optimization problem. We further provide a tractable reformulation for the DRO problem. Second, we prove that the proposed ambiguity set can include

the underlying distribution with a high probability when the sample size and ambiguity aversion level (which determines the size of the ambiguity set) are larger than a quantitative bound. We further analyze the quality of the obtained portfolio from our data-driven approach. Finally, by conducting extensive empirical studies with the datasets in DeMiguel et al. (2009) and a dataset with various asset classes, we show that the portfolio from our approach performs stably and consistently beats the equally weighted portfolio in terms of Sharpe ratio. In addition, we numerically show that our proposed formulation can guide investors to shift their portfolio weightings to less risky assets to optimize their risk profiles, particularly in years such as the 2008-2009 great recession.

The remainder of this paper is organized as follows. Section 2 reviews the related works. Section 3 introduces our distributionally robust portfolio optimization model. Specifically, we formulate the distributionally robust portfolio optimization model under a regime-switching ambiguity set and present its tractable reformulations. In Section 4, we introduce our data-driven portfolio allocation approach. In particular, two methods are proposed to determine the financial regimes and estimate the corresponding transition probabilities. We further analyze the statistical performance guarantees of the proposed data-driven methods. Section 5 conducts numerical studies with extensive empirical datasets to compare the out-of-sample performance of the proposed portfolio with other benchmarks. Finally, Section 6 concludes the paper.

Throughout this paper, we use the following notations. For a vector or a matrix A , the transpose of A is denoted by A' . We denote by $\mathbf{1}$ the vector of ones with appropriate dimensions. For $\mathbf{a} = (a_1, \dots, a_p)' \in \mathbb{R}^p$ and a positive integer m , we denote by ℓ_m -norm $\|\mathbf{a}\|_m = (\sum_{j=1}^p |a_j|^m)^{1/m}$. For an integer M , $[M] := \{1, \dots, M\}$ denotes the index set from 1 to M . Let $\mathcal{P}(\mathcal{H})$ be the space of Borel probability measures supported on $\mathcal{H}$, where $\mathcal{H}$ could be a product space. We denote the $(a, \mathbf{b}) \in \Omega_{soc}$ as the second-order conic constraint, i.e. $\|\mathbf{b}\|_2^2 \leq a^2$. We use δ_x to represent Dirac measure on the single point x and $\mathbb{I}_{\{\cdot\}}$ denote the indicator function. For two distributions μ and ν , we denote $\mu \otimes \nu$ as their product measure. We denote $f(n) = \Theta(g(n))$ such that $\exists k_1, k_2 > 0$ and $\exists n_0$ such that $\forall n > n_0, k_1 g(n) \leq f(n) \leq k_2 g(n)$. Besides, $X_n = O_p(a_n)$ as $n \rightarrow \infty$ means that $\forall \varepsilon > 0, \exists M, N > 0$ such that $\Pr\left(\left|\frac{X_n}{a_n}\right| > M\right) \leq \varepsilon, \forall n > N$.

2. Related Works

In this section, we briefly review the works related to our study from three aspects, namely distributionally robust optimization, optimization under time-varying uncertainties, and regime-switching models.

2.1. Distributionally Robust Optimization

Distributionally robust optimization has received wide attention over the past decades. Different forms of ambiguity sets are developed. Delage and Ye (2010), Wiesemann et al. (2014) examined a moment-based ambiguity set. They provided tractable reformulations for various DRO problems and showed that the model structure of the portfolio optimization admits an analytical solution for its optimal decision. Liu et al. (2019) and Kang et al. (2019) investigated tractable reformulations for portfolio optimization under various moment-based ambiguity sets using the worst-case CVaR as a risk measure. However, using a moment-based ambiguity set may lead to a too-conservative portfolio allocation decision.

Recently, many researchers have focused on studying a statistical-distance-based ambiguity set. Some commonly used distance measures include the ϕ -divergence (Ben-Tal et al. 2013) and Wasserstein distance (Esfahani and Kuhn 2018). We refer readers to a more comprehensive discussion in Rahimian and Mehrotra (2019). Wasserstein ambiguity sets are known to yield a higher chance of nesting the true probability distribution and thus achieve a better out-of-sample performance guarantee. As a result, they are widely used in real applications of portfolio optimization models (see Esfahani and Kuhn 2018, Blanchet et al. 2022). Our paper uses the Wasserstein ambiguity set in designing the portfolio allocation policy.

A recent inspiring work by Chen et al. (2020) provided a generic framework for a large family of ambiguity sets. They propose a *robust stochastic optimization* framework based on a scenario-wise ambiguity set, which is flexible in re-creating various types of ambiguity sets in the literature, including the Wasserstein ambiguity set that is of interest in this paper. Specifically, their scenario-wise ambiguity set is formulated as follows:

$$\bar{\mathcal{U}}(\theta) := \left\{ \mathbb{P} \in \mathcal{P}(\mathbb{R}^I \times [N]) \left| \begin{array}{l} (\tilde{\mathbf{r}}, \tilde{n}) \sim \mathbb{P} \\ \mathbb{E}^{\mathbb{P}}[\tilde{\mathbf{r}} \mid \tilde{n} = n] \in \Omega_n, \forall n \in [N] \\ \mathbb{P}[\tilde{\mathbf{r}} \in \mathcal{Z}_n \mid \tilde{n} = n] = 1, \forall n \in [N] \\ \mathbb{P}[\tilde{n} = n] = p_n, \forall n \in [N] \end{array} \right. \right\} \quad (1)$$

where $\tilde{\mathbf{r}}$ represents a random vector, $\tilde{n}$ indicates a set of random scenarios and p_n denotes the probability of each scenario n . Ω_n is defined as a closed and convex set that can contain some moment conditions, and $\mathcal{Z}_n$ specifies the support of $\tilde{\mathbf{r}}$. They provide several applications of the ambiguity set to show how it recreates the existing ambiguity sets in the literature. In particular, by defining each scenario as an observation, denoted by $\mathbf{r}_n$, and the support of the random vector under each scenario as $\mathcal{Z}_n = \{\tilde{\mathbf{r}} | \rho(\tilde{\mathbf{r}}, \mathbf{r}_n) \leq \theta\}$, where $p(\cdot, \cdot)$ denotes the cost function in the Wasserstein metric, the ambiguity set is reduced to a Wasserstein ambiguity set. This paper builds on their framework and incorporates the regime-switching pattern in the ambiguity set to cope with the nonstationarity.

2.2. Optimization under Time-varying Uncertainty

Traditional DRO models often assume the distribution of uncertainties is time-invariant. However, time-varying uncertainties are a common feature in practice, especially in financial problems. Ignoring the time-varying feature of uncertainties in a decision-making model may deteriorate the solution's (out-of-sample) performance.

Recently, some work has started modeling the uncertainty across time as a dependent structure, specifically through some Markov chains. Dou and Animescu (2019) studied an optimization model where the uncertain parameters are sampled from a special type of time series, vector autoregression process (VAR). They obtained a tractable DRO reformulation for the problem and provided an out-of-sample guarantee for their method. In contrast, our paper models the time-varying uncertainties from a Markov chain perspective. Sutter et al. (2020) and Li et al. (2021) studied an optimization model where the uncertain parameters are sampled from a finite-state Markov chain to model the time-varying behavior. They prove that their data-driven DRO model at any given decision converges almost surely to its true expected loss as the sample size goes to infinity. Moreover, they also showed that the solution from their DRO model can generate the least possible out-of-sample risk among the vast class of approaches. In contrast, the random samples (returns) in our model are governed by a Markov chain (for regimes) rather than directly sampled from a Markov chain, i.e., the random return's distribution is determined by regimes that evolve across time according to a Markov chain. In addition, we can establish a finite-sample bound for the ambiguity level to cover the true distribution with a high probability under mild assumptions. We also theoretically analyze the quality of the obtained solution from our data-driven DRO model. Yang (2017) and Ramani and Ghate (2022) modeled the

time-varying uncertainties in a Markov decision process with state-dependent rewards. States evolve over time; hence, the (uncertain) rewards' distributions vary across time. This problem setting is similar to ours but in a multi-stage setting. However, their estimation of the ambiguity set requires multiple i.i.d. sample trajectories. Nevertheless, in many data sets, we can only observe one trajectory of historical data. In summary, we model the time-varying uncertainties by regime-dependent returns, where the regime evolves over time according to a Markov chain. We provide a tractable reformulation for our DRO model with time-varying uncertainties and establish a non-asymptotic performance guarantee for the proposed data-driven approach.

Another related stream of literature is to study the estimation of uncertainties from exogenous covariates (see Bertsimas and Van Parys 2021, Esteban-Pérez and Morales 2021). They assume the covariates are exogenous and i.i.d. across time. The uncertainties' distribution could vary depending on different realizations of the covariates, which leads to time-varying uncertainties. In contrast, we assume the covariates are sampled from a Markov chain rather than i.i.d across time. In particular, we use the market state and macroeconomic variables as covariates in predicting regimes that determine the returns' distribution.

In summary, this paper contributes to the optimization under time-varying uncertainties from two perspectives. From the modeling perspective, we exploit one specific type of non-stationarity of the time-series data by assuming that the uncertainties (e.g., uncertain returns) generating the data depend on each regime, which may switch from one to another across time according to a Markov chain. We then incorporate the evolution of latent uncertainty sources (such as macro-economics variables) when using them as covariates to predict the distribution of the uncertain returns in different regimes. From the theoretical perspective, we show that our proposed ambiguity sets can include the underlying distribution of interest with high probability and enjoy tractable reformulations and practical interpretations.

2.3. Regime Switching Models

In the finance literature, regime-switching models have been studied in modeling stock returns in the context of economic modeling (Hamilton 1989, 2010). Evidenced by the economic observations (see Tu 2010, Chang et al. 2017), the parameters of the return process are regime-switching. Tu (2010) investigated the economic value of accounting for

regimes in making portfolio allocation decisions and pointed out that ignoring the regime-switching feature will lead to significant certainty-equivalent losses. Subsequent studies examined how to optimize the portfolio allocation decision in a regime-switching market and demonstrated the improvement made by incorporating the regime-switching behavior (Bae et al. 2014). Many variants of portfolio models have also been investigated under the regime-switching pattern by either directly observing the history states or inferring states through HMM (Guidolin and Timmermann 2007, Fu et al. 2014, Costa and Kwon 2019, 2020, Nystrup et al. 2017, Oprisor and Kwon 2021). However, most of them directly use the estimated parameters from the regime-switching model as input in their portfolio optimization model, which does not consider the estimation error of these parameters with finite samples. In contrast, we introduce a variant of Wasserstein ambiguity sets to capture the regime-switching pattern and theoretically analyze the effect of estimation errors on the portfolio's performance. In particular, we show that our constructed ambiguity set with a certain ambiguity aversion level can contain the true distribution of interest with a high probability under mild assumptions. Under the same ambiguity aversion level, we can bound the risk of the obtained portfolio. To the best of our knowledge, this paper is the first to examine portfolio optimization in a regime-switching market from the DRO perspective.

3. Distributionally Robust Portfolio Optimization with Regime-Switching Ambiguity

In this section, we first model the distributionally robust portfolio optimization problem and introduce our regime-switching ambiguity set. Next, we provide a tractable reformulation of the DRO model and discuss the relation between our optimal solution and the equally weighted portfolio—a widely studied portfolio in the literature.

3.1. Preliminaries

3.1.1. Wasserstein Ambiguity Set The Wasserstein ambiguity set is representable by restricting the probability distributions to a certain statistical distance of a reference distribution, such as the empirical probability distribution (denoted by $\hat{\mathbb{P}}_e$). To define a classical Wasserstein ambiguity set, we first introduce the following definitions. For a lower-semi-continuous cost function $\rho: \mathbb{R}^I \times \mathbb{R}^I \mapsto [0, \infty)$ with $\rho(\mathbf{u}, \mathbf{u}) = 0$ for any $\mathbf{u} \in \mathbb{R}^I$, the Wasserstein distance between two probability distributions $\mathbb{P}$ and $\mathbb{Q}$ is defined as:

$$W(\mathbb{P}, \mathbb{Q}) := \inf_{\nu} \left\{ \mathbb{E}^{\nu}[\rho(\mathbf{p}, \mathbf{q})] \mid \nu \in \mathcal{P}(\mathbb{R}^I \times \mathbb{R}^I), \Pi_{\mathbf{p}}\nu = \mathbb{P}, \Pi_{\mathbf{q}}\nu = \mathbb{Q} \right\}, \quad (2)$$

where ν is the joint distribution of $\mathbb{P}$ and $\mathbb{Q}$ and $\Pi_{\mathbf{p}}\nu$ and $\Pi_{\mathbf{q}}\nu$ are the projections of ν over $\mathbf{p}$ and $\mathbf{q}$, respectively. The metric (2) refers to the optimal transportation cost of moving the mass from $\mathbb{P}$ into the mass of $\mathbb{Q}$ under a cost $\rho(\mathbf{p}, \mathbf{q})$ per unit of mass transported from location $\mathbf{p}$ to location $\mathbf{q}$. This paper considers the cost function as ℓ_m -norm: $\rho(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\|_m$ and in this case, (2) is known as the Wasserstein distance with ℓ_m -norm and we denote it by W_m . Define the ambiguity set as,

$$\mathcal{U}_w(\theta) = \left\{ \mathbb{P} \in \mathcal{P}(\mathbb{R}^I) \mid W_m(\mathbb{P}, \hat{\mathbb{P}}) \leq \theta \right\}, \quad (3)$$

which is a Wasserstein ambiguity set of radius θ centered at the reference distribution $\hat{\mathbb{P}}$. Along this line of work, the distinct feature of our paper is the incorporation of regime-switching behavior arising from the investing environment.

3.1.2. CVaR Minimization Formula Consider a market with I risky assets whose random return vector is represented by $\mathbf{r} = (r_1, \dots, r_I)'$ for the specified investment horizon. The portfolio of the investor is characterized by the portfolio weight vector $\mathbf{x} = (x_1, \dots, x_I)'$ with $\mathbf{1}'\mathbf{x} = 1$, where x_i is the proportion of the investor wealth that is invested in the i -th asset at the beginning of the investment for $i = 1, \dots, I$. In our framework, we consider the feasible portfolios that satisfy the general linear equality and inequality constraints, i.e.,

$$\mathbf{x} \in \mathcal{X} := \{\mathbf{x} \mid A\mathbf{x} = \mathbf{b}, C\mathbf{x} \geq \mathbf{d}\},$$

where $A \in \mathbb{R}^{m \times I}$, $\mathbf{b} \in \mathbb{R}^m$, $C \in \mathbb{R}^{p \times I}$, and $\mathbf{d} \in \mathbb{R}^p$. Some realistic portfolio constraints, such as leverage or no-shorting constraints, can be incorporated into our framework. For notation simplicity, we embed the budget constraint $\mathbf{1}'\mathbf{x} = 1$ into $\mathcal{X}$ as a necessary constraint.

Given the portfolio $\mathbf{x}$, the portfolio's return is equal to $\mathbf{x}'\mathbf{r}$. Thus, we define the loss (function) of the investor as $f(\mathbf{x}, \mathbf{r}) := -\mathbf{x}'\mathbf{r}$, name it as $\mathbb{P}$. Assume that the random return vector $\mathbf{r}$ has a probability density function, $p(\mathbf{r})$. For a portfolio weight vector $\mathbf{x}$ and a constant v , the cumulative distribution function of the loss $f(\mathbf{x}, \mathbf{r})$ is denoted by $\psi(\mathbf{x}, v) := \int_{f(\mathbf{x}, \mathbf{r}) \leq v} p(\mathbf{r}) d\mathbf{r}$. For a confidence level $\eta \in (0, 1)$, VaR is defined as $\text{VaR}_\eta(\mathbf{x}) = \inf\{v \in \mathbb{R} \mid \psi(\mathbf{x}, v) \geq \eta\}$ and the corresponding CVaR is defined as the conditional expectation of the loss $f(\mathbf{x}, \mathbf{r})$ greater than VaR_η :

$$\text{CVaR}_\eta^\mathbb{P}(\mathbf{x}) = \mathbb{E}^\mathbb{P}[f(\mathbf{x}, \mathbf{r}) \mid f(\mathbf{x}, \mathbf{r}) \geq \text{VaR}_\eta(\mathbf{x})] = \frac{1}{(1 - \eta)} \int_{f(\mathbf{x}, \mathbf{r}) \geq \text{VaR}_\eta(\mathbf{x})} f(\mathbf{x}, \mathbf{r}) p(\mathbf{r}) d\mathbf{r}. \quad (4)$$

As shown in Rockafellar and Uryasev (2002), the $\text{VaR}_\eta(\mathbf{x})$ and $\text{CVaR}_\eta(\mathbf{x})$ are the minimizer and the minimum value of a function $F_\eta(\mathbf{x}, v)$ with respect to v under $\mathbf{r} \sim \mathbb{P}$, respectively, where

$$F_\eta^\mathbb{P}(\mathbf{x}, v) := v + \frac{1}{1-\eta} \mathbb{E}^\mathbb{P}[(f(\mathbf{x}, \mathbf{r}) - v)^+],$$

in which $g^+ := \max\{g, 0\}$ is the positive part of g . It is shown that $F_\eta(\mathbf{x}, v)$ is convex in $\mathbf{x}$ and v .

The stochastic CVaR portfolio optimization can be formulated as follows:

$$\min_{\mathbf{x} \in \mathcal{X}} \text{CVaR}_\eta^\mathbb{P}(\mathbf{x}) = \min_{\mathbf{x} \in \mathcal{X}, v \in \mathbb{R}} F_\eta^\mathbb{P}(\mathbf{x}, v) = \min_{\mathbf{x} \in \mathcal{X}, v \in \mathbb{R}} \left\{ v + \frac{1}{1-\eta} \mathbb{E}^\mathbb{P}[G(\mathbf{x}, \mathbf{r})] \right\}, \quad (5)$$

where

$$G(\mathbf{x}, \mathbf{r}) = \min_{y \in \mathcal{Y}} y \quad \text{with } \mathcal{Y} = \{y \mid y \geq -\mathbf{x}'\mathbf{r} - v, y \geq 0\}. \quad (6)$$

Given that the feasible set $\mathcal{X}$ is convex, the solution to (5) can be obtained by standard convex optimization algorithms (see Krokhmal et al. 2001).

3.2. Regime-Switching Ambiguity Set

In practice, the computation of CVaR or the expectation in (5) requires the knowledge of the return distribution. Traditionally, by using the historical return data of the length N , $\{\mathbf{r}_1, \dots, \mathbf{r}_N\}$, one may adopt the following empirical version of (5) as its proxy:

$$\min_{\mathbf{x} \in \mathcal{X}, v \in \mathbb{R}} \left\{ v + \frac{1}{N(1-\eta)} \sum_{n=1}^N [G(\mathbf{x}, \mathbf{r}_n)] \right\}, \quad (7)$$

Essentially, in (7), the return distribution is estimated by the empirical probability distribution, denoted by $\hat{\mathbb{P}}_e := \frac{1}{N} \sum_{n \in [N]} \delta_{\mathbf{r}_n}$, where $\delta_{\mathbf{r}_n}$ denotes a Dirac delta function placed at a location $\mathbf{r}_n \in \mathbb{R}^I$. However, this estimation procedure suffers from considerable errors, leading to the resulting portfolio's inferior empirical performance.

A popular remedy is to apply a DRO approach. Zhu and Fukushima (2009) studied the CVaR under the worst-case distribution among a family of distributions named ambiguity set. They show that worst-case expected CVaR inherits the properties of CVaR, including subadditivity, positive homogeneity, monotonicity, and translation invariance. The corresponding robust CVaR portfolio optimization is formulated as follows:

$$\min_{\mathbf{x} \in \mathcal{X}, v \in \mathbb{R}} \left\{ v + \frac{1}{1-\eta} \sup_{\mathbb{P} \in \mathcal{U}} \mathbb{E}^\mathbb{P}[G(\mathbf{x}, \mathbf{r})] \right\} \quad (8)$$

where $\mathcal{U}$ is the ambiguity set of return distributions. Note that problem (8) is a two-stage optimization problem, which is generally difficult to solve, but the CVaR metric here allows tractable reformulations (Zhu and Fukushima 2009, Hanasusanto et al. 2015, Long et al. 2021).

REMARK 1. If we define $\mathcal{U} = \mathcal{U}_w(\theta)$ in (3), then problem (8) is a distributionally robust CVaR portfolio optimization under a Wasserstein ambiguity set, where the radius θ represents the level of ambiguity aversion of the investor. It is easy to see that when $\theta = \mathbf{0}$, this robust CVaR portfolio optimization is reduced to (7) since $\mathbb{P} \equiv \hat{\mathbb{P}}_e$ in this case.

Due to the existence of time-varying economic conditions in the financial market, the ambiguity of asset returns is regime-dependent, where each regime refers to an economic cycle. It is crucial to incorporate the prediction of economic conditions in constructing an ambiguity set. In our ambiguity set, we define each economic condition as a regime. Conditioning on a regime, we model the random return in a Wasserstein ambiguity set defined under Chen et al. (2020). Specifically, the regime-switching Wasserstein ambiguity set is defined as follows:

$$\mathcal{U}_{rs}(\theta) = \left\{ \mathbb{P} \in \mathcal{P} \left(\mathbb{R}^I \times \mathbb{R}^K \times [N_k]_{k=1}^K \times [K] \right) \left| \begin{array}{l} (\tilde{\mathbf{r}}, \tilde{\mathbf{u}}, \tilde{n}, \tilde{k}) \sim \mathbb{P} \\ \mathbb{E}^{\mathbb{P}} [\tilde{u}_k] \leq \theta_k, \forall k \in [K] \\ \mathbb{P} [\tilde{\mathbf{r}} \in \mathbb{R}^I, \rho(\tilde{\mathbf{r}}, \mathbf{r}_{nk}) \leq \tilde{u}_k \mid n \in [N_k], k \in [K]] = 1 \\ \mathbb{P} [\tilde{n} = n \mid \tilde{k} = k] = \frac{1}{N_k}, \forall n \in [N_k], k \in [K] \\ \mathbb{P} [\tilde{k} = k] = w_k, \forall k \in [K] \end{array} \right. \right\}, \quad (9)$$

where there are K different regimes, $\tilde{k}$ denotes the regime (e.g., the economic condition) from which the uncertain return is drawn, and takes value in $[K]$. In each regime k , there are N_k return samples ($\mathbf{r}_{nk}$ for $k \in [N_k]$), which provide an empirical distribution of returns $\hat{\mathbb{P}}_k := \frac{1}{N_k} \sum_{n \in [N_k]} \delta_{\mathbf{r}_{nk}}$. We use $\tilde{n}$ to represent a discrete random scenario of returns which takes values in $[N_k]$ under each regime k . The sample of return in scenario n under regime k is $\mathbf{r}_{nk}$. $\rho(\cdot, \cdot)$ denotes the cost function in the Wasserstein metric (2). θ_k measures the ambiguity aversion level under each regime k . Following a traditional approach (see Wiesemann et al. 2014, Chen et al. 2020), we introduce a vector of K auxiliary random variables $\tilde{\mathbf{u}} = (\tilde{u}_1, \dots, \tilde{u}_K)'$ in the ambiguity set for the sake of modeling tractability. w_k denotes the probability that a random return is from regime k . We will provide different ways to determine these parameters in Section 4.

The essence of the proposed ambiguity set is to restrict the distribution within a pre-specified threshold θ_k from the empirical distribution in each regime. A natural relaxation is that we can relax the constraint of the distribution to a set of distributions located within an average threshold from the average empirical distribution, where the average is taken across different regimes. This relaxation helps apply the existing methods in the literature of Wasserstein distance to optimize (8). However, we show in Proposition 1 that the proposed ambiguity set (projected to the random returns) is a strict subset of the traditional Wasserstein ambiguity set that bounds the distance to the average empirical distribution by the average threshold. In other words, it is not straightforward to apply the existing methods in the literature of the Wasserstein ambiguity set to reformulate (8) with the ambiguity set $\mathcal{U}_{rs}(\boldsymbol{\theta})$ in (9).

PROPOSITION 1. *The regime-switching ambiguity set is a strict subset of the Wasserstein ambiguity set with the distance upper bounded by the weighted average ambiguity aversion level $\sum_{k \in [K]} w_k \theta_k$, i.e.,*

$$\Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}(\boldsymbol{\theta}) \subsetneq \mathcal{U}_w\left(\sum_{k \in [K]} w_k \theta_k\right) = \left\{ \mathbb{P} \in \mathcal{P}(\mathbb{R}^I) \mid W_m(\mathbb{P}, \hat{\mathbb{P}}) \leq \sum_{k \in [K]} w_k \theta_k \right\}, \quad (10)$$

where $\hat{\mathbb{P}} = \sum_{k \in [K]} \frac{w_k}{N_k} \sum_{n \in [N_k]} \delta_{\mathbf{r}_{nk}}$ and $\Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}(\boldsymbol{\theta})$ is the projection on the first coordinate that corresponds to $\tilde{\mathbf{r}}$. Specifically, it represents a set of $\tilde{\mathbf{r}}$ where there exists some $(\tilde{\mathbf{u}}, \tilde{n}, \tilde{k})$ such that $(\tilde{\mathbf{r}}, \tilde{\mathbf{u}}, \tilde{n}, \tilde{k}) \in \mathcal{U}_{rs}(\boldsymbol{\theta})$.

REMARK 2. If there is only one regime ($K = 1$, $k = \tilde{k} = w_k \equiv 1$, and $N_k = N$), (9) is reduced to (1) in the application of the Wasserstein ambiguity set according to Chen et al. (2020). The essential difference between (9) and (1) is that we introduce a regime random variable $\tilde{k}$ which transits from one to another according to a Markov chain in (9) to characterize the regime-switching ambiguity. (9) represents K different Wasserstein ambiguity sets mixed with the time-varying (state-dependent) transition probabilities.

In summary, the key contribution of this part is to refine the representation of the ambiguity set initiated by Chen et al. (2020). The refined ambiguity set helps incorporate the financial intuitions into the modeling framework. But it also brings challenges in computing the optimal portfolio allocation. In the next subsection, we will provide a reformulation of the DRO problem (8) under a regime-switching ambiguity set and conditions of when the problem remains tractable.

3.3. Reformulation of Distributionally Robust Portfolio Optimization with Regime-Switching Ambiguity

The proposed portfolio is found by solving the problem (8) with $\mathcal{U} = \mathcal{U}_{rs}(\boldsymbol{\theta})$:

$$\min_{\mathbf{x} \in \mathcal{X}, v \in \mathbb{R}} \left\{ v + \frac{1}{1-\eta} \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \mathbb{E}^{\mathbb{P}}[G(\mathbf{x}, \mathbf{r})] \right\}, \quad (11)$$

where G is given in (6). The following proposition shows that the distributionally robust CVaR portfolio optimization with the regime-switching ambiguity set where the Wasserstein distance is defined in ℓ_m -norm ($m \geq 1$) can be reformulated as a convex optimization problem.

THEOREM 1. *$\forall m \in [1, +\infty]$, the problem (11) under the ambiguity set $\mathcal{U}_{rs}(\boldsymbol{\theta})$ in (9) with $\rho(\cdot, \cdot)$ defined by ℓ_m -norm can be reformulated as:*

$$\begin{aligned} \min_{v, \mathbf{x}, \boldsymbol{\alpha}, \beta} \quad & v + \frac{1}{1-\eta} \left[\sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \alpha_{nk} + \beta \sum_{k \in [K]} w_k \theta_k \right] \\ \text{s.t.} \quad & \mathbf{r}'_{nk} \mathbf{x} \geq -\alpha_{nk} - v, \forall n \in [N_k], k \in [K], \\ & \beta \geq \|\mathbf{x}\|_{m^*}, \\ & \alpha_{nk} \geq 0, \forall n \in [N_k], k \in [K], \\ & \mathbf{x} \in \mathcal{X}. \end{aligned} \quad (12)$$

In particular, when $m = 1$ and $m = 2$, (12) is a linear program and second-order conic program, respectively, which are both tractable.

In practice, we can incorporate the target on the empirical mean return in the constraint set $\mathcal{X}$ without affecting the tractability. This is because the formula of the empirical mean return is a linear function of the portfolio allocation decision $\mathbf{x}$. Specifically, the constraint on the target portfolio's return is $\boldsymbol{\mu}' \mathbf{x} \geq R$, where $\boldsymbol{\mu}$ is the anticipated mean return vector, which can be estimated from data, and R is the targeted return (see Yao et al. 2013). An alternative way to consider the target of mean returns in the literature is the robust Mean-CVaR model, where the worst case expected return is upper bounded by a target. We provide a tractable DRO formulation for the robust Mean-CVaR model in Appendix EC.3.

Finally, we would like to comment on the solution from our model. Note that a popular portfolio allocation policy studied in the literature is to equally allocate portfolios. We show in Proposition 2 that our model would converge to this classical equally weighted portfolio when the radius of our ambiguity set is large enough.

PROPOSITION 2. *When $\sum_k w_k \theta_k \rightarrow \infty$, the solution of the DRO problem (8) with ambiguity set $\mathcal{U}_{rs}(\boldsymbol{\theta}) \forall m \in [1, +\infty]$ converges to the equally weighted portfolio.*

4. Data-Driven Portfolio Allocation Approach: Algorithms and Theoretical Analysis

The key to determining our ambiguity set is to determine the regimes and estimate the transition probabilities between every pair of them. In this section, we will provide two algorithms to estimate regimes and transition probabilities. This section is organized as follows. First, we will provide an overall procedure to construct the optimal portfolio from historical data, where two different approaches are introduced to determine regimes and estimate the associated transition probabilities. Then we theoretically analyze the performance of the constructed portfolios. Before proceeding, we first list the following assumption on the regimes.

ASSUMPTION 1 (**Markov Chain**). *The sequences of regimes $\{s_i\}_{i \in \mathbb{Z}}$ form a time-homogeneous ergodic Markov Chain in $[K]$ with a unique stationary distribution $(\pi_1, \dots, \pi_K)$. The conditional distribution of the true return $\tilde{\mathbf{r}}|s=i, \forall i \in [K]$ exhibits variations across different regime s .*

4.1. Algorithms

We present our data-driven portfolio allocation algorithm in Algorithm 1, with specific blocks in Algorithms 2, 3, and 4 in Appendix EC.4. Here the index n of $\mathbf{r}$, s , and o represents the return, regime, and output observation at time n , respectively. We denote the constructed portfolio weight from LP (12) as $\mathbf{x}^*(\boldsymbol{\theta}; \mathcal{R}_t, s)$ given the regime s under the training sample $\mathcal{R}_t$ with the ambiguity aversion level $\boldsymbol{\theta}$.

This algorithm applies the “rolling-sample” approach proposed by DeMiguel et al. (2009). Specifically, given a T -month-long data, in each month t , we use the dataset from months t to $t+N-1$ to estimate the ambiguity set $\mathcal{U}_{rs}(\boldsymbol{\theta})$ (see (9)) and solve the portfolio optimization problem (see (12)), which generates the portfolio weight and the total return in month $t+N$. We repeat the process by adding the returns for the next period and dropping the earliest one until we reach the end of the dataset.

In estimating the ambiguity set $\mathcal{U}_{rs}(\boldsymbol{\theta})$, several parameters need to be determined, which include each regime, its associated weight, and the ambiguity aversion level $\boldsymbol{\theta}$. We apply two

Algorithm 1 Data-driven Regime-Switching Portfolio Selection Procedure

Input: Window size N , whole time period with returns $M(\leq T)$, sequences of asset returns $\{\mathbf{r}_1, \dots, \mathbf{r}_M\}$, where $\mathbf{r}_i \in \mathbb{R}^I, \forall i \in [M]$; estimated regimes (input of Algorithm 2) or observational sequences (input of Algorithm 3), set of candidate aversion level Θ (input of Algorithm 4).

- 1: **for** $t = 1, \dots, M - N$ **do**
- 2: Obtain the training sample of asset returns $\mathcal{R}_t := \{\mathbf{r}_t, \dots, \mathbf{r}_{N+t-1}\}$ by the “rolling-sample” approach;
- 3: Determine the $\{w_k, \{\mathbf{r}_{nk}\}_{n \in [N_k]}\}_{k \in [K]}$ in the ambiguity set (9) from $\mathcal{R}_t$ using the frequency-based approach in Algorithm 2 or the hidden Markov approach in Algorithm 3.
- 4: Cross-validation to determine the best ambiguity aversion level θ^* from Θ (see Algorithm 4 in Appendix EC.4).
- 5: Solve the optimization problem in (12) to obtain the optimal portfolio weight vector $\mathbf{x}^*(\theta^*; \mathcal{R}_t, s_{N+t-1})$, denoted by $\mathbf{x}_{N+t}^*$ for simplicity. Then the realized total return denoted by $\hat{g}_{N+t}$ is computed by $\hat{g}_{N+t} = \mathbf{r}'_{N+t} \mathbf{x}_{N+t}^* \in \mathbb{R}$.
- 6: **end for**

Output: constructed portfolio weight $\{\mathbf{x}_{N+1}^*, \dots, \mathbf{x}_M^*\}$; the associated total return sequence $\{\hat{g}_{N+1}, \dots, \hat{g}_M\}$.

different approaches named a frequency-based approach (see Section 4.1.1) and a hidden-Markov approach (see Section 4.1.2) to estimate regimes and their weights, and use cross-validation to tune θ . The detailed procedure for cross-validation is shown in Algorithm 4 in Appendix EC.4. Different from the standard m -fold cross-validation procedure (i.e. Ban et al. 2018), we cannot split samples randomly since the distribution in the subsequent periods depend on the regime at the current time. We also cannot validate $\mathcal{D}_{t,1}$ using the training data $\mathcal{D}_t \setminus \mathcal{D}_{t,1}$ since the validating step requires s_{t-1} , which is beyond the input in our implementation. To resolve these issues, we split samples in the temporal order and validate from the second fold (start from $i = 2$ instead of $i = 1$ in Step 2 of Algorithm 4 in Appendix EC.4).

4.1.1. A Frequency-based Approach. This section introduces several common financial practices in determining regimes and provides estimation methods for transition probabilities.

A simple yet widely used definition of the financial market regimes is bull and bear markets (see Tu 2010). These two regimes are characterized by the market’s trend. If the

market return tends to be positive, it is defined as a bull market; otherwise, it is a bear market. The market return can be measured by the market portfolio (capitalized weighted index) or equally-weighted portfolio of all the assets in the market. In our experiments, we stick with the latter because some assets of our choice are sector portfolios. We can compute the market return for bull and bear markets and determine the corresponding regime at each time point.

When there are different asset classes in the market, the bull and bear markets appear to be insufficient in describing the market regimes. Inspired by the success of the *All Weather* fund launched by Bridgewater Associates in 1996, one can determine the financial regimes using some macroeconomic variables. For example, bonds perform best during times of deflationary recession; stocks do well during periods of growth; holding cash becomes the most attractive option when the money supply is tight. By investigating market dynamics and potential return scenarios over a long period, Bridgewater (2012) point out that economic drivers, such as inflation and growth rates, can determine which asset class an investor should invest in. Therefore, it is potentially interesting to incorporate this regime-switching nature of assets in estimating the regimes. Figure 1 presents the categorization of the regimes by Bridgewater and the asset classes that perform well in each regime. For

		Growth	Inflation
Market Expectations	Rising	Equities Commodities Corporate Credit EM Credit	IL Bonds Commodities EM Credit
	Falling	Nominal Bonds IL Bonds	Equities Nominal Bonds

Figure 1 The weathers (regimes) defined in Bridgewater (2012) and in the boxes, the asset classes would rise over time. IL stands for inflation-linked and EM stands for emerging market.

the four regimes defined in Bridgewater (2012), we can classify the data into four different economic scenarios using the 2×2 interactions of US CPI and GDP data that represent the levels of inflation and growth, respectively. Specifically, we determine if the inflation is rising above or falling below the market's expectations by subtracting the moving average of the recent one-year US CPI from its current value. Similarly, we do the same for the growth using the US GDP data.

Algorithm 2 Frequency-based Approach

Input: Return sequences $\mathcal{R}_t$, associated regime sequences $\{s_t, \dots, s_{t+N-1}\}$ determined in Section 4.1.1.

- 1: Obtain the parameters of transition probability matrix by $a_{j,k} = \frac{\sum_{i=t}^{t+N-2} \mathbb{I}_{\{s_i=j, s_{i+1}=k\}}}{\sum_{i=t}^{t+N-2} \mathbb{I}_{\{s_i=j\}}}, \forall j, k \in [K]$.
- 2: From the last **observed** regime $j^* := s_{N+t-1}$, determine the regime frequency by $w_k = a_{j^*,k}$ and associated return sequences by $\{\mathbf{r}_{nk}\}_{n \in [N_k]} = \{\mathbf{r}_i | s_i = k, t \leq i \leq N+t-1\}, \forall k \in [K]$ for (9).

Output: Transition probability matrix $A = \{a_{j,k}\}_{j,k \in [K]}$; $\{w_k, \{\mathbf{r}_{nk}\}_{n \in [N_k]}\}_{k \in [K]}$ in the ambiguity set (9).

After determining the regimes, we can estimate the transition probability matrix $A = \{a_{j,k}\}_{j,k \in [K]}$ for the Markov chain by the frequency of occurrence from one regime to another. Note that $\sum_{k \in [K]} a_{j,k} = 1$ for all $j \in [K]$. The general procedure of the frequency-based approach is shown in Algorithm 2. We also illustrate how our algorithm determines the parameters in (9) in each period using Example 1. Here we denote the transition probability matrix of the true regime sequence to be $A^* = \{a_{j,k}^*\}_{j,k \in [K]}$.

EXAMPLE 1. Suppose we have 10 sequential history samples of asset returns $\{\mathbf{r}_1, \dots, \mathbf{r}_{10}\}$ and there are two underlying regimes 1 and 2. $\tilde{\mathbf{r}}|s=1 \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\tilde{\mathbf{r}}|s=2 \sim \mathcal{N}(-\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The underlying true regime is $\mathbf{s}^* = \{1, 2, 2, 1, 1, 2, 1, 2, 2, 1\}$ with regime transition matrix $A^* = \begin{pmatrix} 0.2 & 0.8 \\ 0.6 & 0.4 \end{pmatrix}$. We would like to model the ambiguity set of $\tilde{\mathbf{r}}_{11}$ from data. Since $s_{10} = 1$, the underlying true distribution of our interest $\tilde{\mathbf{r}}_{11} \sim 0.2\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) + 0.8\mathcal{N}(-\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Using the frequency-based approach, given the input $\{\mathbf{r}_i\}_{i \in [10]}$ and one observed regime sequence: $\mathbf{s} = \{1, 2, 1, 1, 1, 2, 2, 1, 2, 1\}$, we apply Step 1 of Algorithm 2 to estimate the transition matrix $A = \begin{pmatrix} 0.4 & 0.6 \\ 0.75 & 0.25 \end{pmatrix}$. Since the last observed state is $s_{10} = 1$, we output $w_1 = a_{1,1} = 0.4, w_2 = a_{1,2} = 0.6, N_1 = 6, N_2 = 4, \{\mathbf{r}_{n1}\}_{n \in [N_1]} = \{\mathbf{r}_i | s_i = 1\} = \{\mathbf{r}_1, \mathbf{r}_3, \mathbf{r}_4, \mathbf{r}_5, \mathbf{r}_8, \mathbf{r}_{10}\}, \{\mathbf{r}_{n2}\}_{n \in [N_2]} = \{\mathbf{r}_i | s_i = 2\} = \{\mathbf{r}_2, \mathbf{r}_6, \mathbf{r}_7, \mathbf{r}_9\}$ according to Step 2 of Algorithm 2, which determines the parameters of the ambiguity set in (9).

4.1.2. A Hidden-Markov Approach The frequency-based method requires us to know the exact regime of the economy at any point in time, which may not be directly observable and representative of the asset return regime. Therefore, we propose a hidden-Markov model (HMM) to determine the regimes and the corresponding transition probability matrix simultaneously; see Ghahramani (2001) for an introduction to HMM and

Stratonovich (1960), Baum and Petrie (1966) for the mathematics behind HMM. Note that even though the regimes can be hidden, we still need to specify the number of the regimes using this approach.

Following Assumption 1, we assume there are K regimes in the financial market. We use $O = \{o_1, \dots, o_T\}$ to denote a sequence of T observations. The definition of each observation is based on the available information in the data set. In Section 5.2, Appendix EC.6.1, when dealing with the classical DeMiguel et al. (2009)'s datasets, the historical return data are available, we use $o_t \in \{+, -\}$ to denote whether the market returns are above the risk-free rate (given by the 90-day Treasury-bill yield) at time t . In Section 5.3, there are five different asset classes in the data: equities, treasury bonds, commodities, credit, and treasury inflation-protected securities (TIPS), indexed from $i = 1$ to $i = 5$. Then we define $o_t = i$ if the asset class i enjoys the highest return among all five asset classes at time t .

We use $\mathbb{P}^{HMM}$ to denote the probability distribution of the sequence of observations and the corresponding regimes. An HMM instantiates two simplifying assumptions:

Markov assumption. The probability of a particular regime depends only on the previous regime:

$$\mathbb{P}^{HMM}(s_t | s_{t-1}, s_{t-2}, \dots, s_1) = \mathbb{P}^{HMM}(s_t | s_{t-1}).$$

Output independence. The probability of an output observation o_t depends only on the regime s_t that produces the observation and not on any other regimes or any other observations:

$$\mathbb{P}^{HMM}(o_t | s_1, \dots, s_T, o_1, \dots, o_T) = \mathbb{P}^{HMM}(o_t | s_t).$$

Given a sequence of observations O , the HMM can tackle the following two tasks:

1. Provided the inputs of transition probability matrix $A = \{a_{j,k}\}_{j,k \in [K]}$ and a sequence of observation likelihoods (emission probabilities), denoted by $B = \{b_{i,j}\}_{i \in [K]}^{j \in \mathcal{O}}$, where $b_{i,j} = \Pr(o = j | s = i)$, determine $\mathbb{P}^{HMM}(O | A, B)$ and the most likely hidden regime sequence $(s_1, \dots, s_T)$ corresponding to the observations.

2. Provided a given regime sequence, learn the HMM parameters, A and B .

The standard algorithm for training HMM is the Viterbi algorithm for the first task (Viterbi 1967, Forney 1973) and the forward-backward algorithm (Baum 1972), which is a special case of Expectation-Maximization (EM) algorithm (Dempster et al. 1977), for the second task. The combined algorithm first computes an initial estimate for the

regimes and the transition probabilities and iteratively improves $\mathbb{P}^{HMM}(O|A, B)$. Readers are referred to the appendix of Jurafsky and Martin (2008) for a tutorial on training HMM. We denote the combined algorithm as one **HMM training oracle** with only a sequence of observations O as input. The output of the **HMM training oracle** includes its most likely regime assignment for the observations and the transition probability matrix. The specific procedure of HMM is shown in Algorithm 3.

Algorithm 3 HMM Approach

Input: Return sequences $\mathcal{R}_t$, associated observation sequences $\{o_t, \dots, o_{t+N-1}\}$ determined in Section 4.1.2

- 1: Obtain the parameters of transition probability matrix $A = \{a_{j,k}\}_{j,k \in [K]}$ and most likely regimes $\{s_t, \dots, s_{t+N-1}\}$ from one **HMM training oracle**.
- 2: From the last **predicted** regime $j^* := s_{N+t-1}$ the step above, determine the regime frequency by $w_k = a_{j^*,k}$ and associated return sequences by $\{\mathbf{r}_{nk}\}_{n \in [N_k]} = \{\mathbf{r}_i | s_i = k, t \leq i \leq N+t-1\}, \forall k \in [K]$ for (9).

Output: Transition probability matrix $A = \{a_{j,k}\}_{j,k \in [K]}$; most likely regimes $\{s_t, \dots, s_{t+N-1}\}$; $\{w_k, \{\mathbf{r}_{nk}\}_{n \in [N_k]}\}_{k \in [K]}$ in the ambiguity set (9).

Following the earlier Example 1, suppose the output emission probability matrix $B = \begin{pmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{pmatrix}$. Using the HMM approach, given the input $\{\mathbf{r}_i\}_{i \in [10]}$ and one observation sequence: $\mathbf{o} = \{-, +, +, -, +, +, -, -, +, -\}$ following the first definition above, we output the estimated transition matrix $A = \begin{pmatrix} 0.25 & 0.75 \\ 0.6 & 0.4 \end{pmatrix}$ and the most likely regimes $\mathbf{s} = \{1, 2, 2, 1, 2, 2, 1, 1, 2, 1\}$ by calling the **hmm** training oracle in Python¹ according to Step 1 of Algorithm 3. Since the last predicted regime $s_{10} = 1$, we output $w_1 = a_{1,1} = 0.25, w_2 = a_{1,2} = 0.75, N_1 = 5, N_2 = 5, \{\mathbf{r}_{n1}\}_{n \in [N_1]} = \{\mathbf{r}_i | s_i = 1\} = \{\mathbf{r}_1, \mathbf{r}_4, \mathbf{r}_7, \mathbf{r}_8, \mathbf{r}_{10}\}, \{\mathbf{r}_{n2}\}_{n \in [N_2]} = \{\mathbf{r}_i | s_i = 2\} = \{\mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_5, \mathbf{r}_6, \mathbf{r}_9\}$ for (9) according to Step 2 of Algorithm 3.

Note that no matter which approach is used, the transition probability matrix is dynamic and will change across time based on the historical data, following the initiative of Bazzi et al. (2016). The length of observations can be different from that of the return data since the specification of the regimes is more of a long-term feature, while the historical return

¹ The training oracle is implemented using the code `hmm.GaussianHMM(2, 'full')` under Python 3.

data are used to infer the returns in the near future. In our case study, we use the ten-year data to determine the regimes.

4.2. Theoretical Analysis of Regime Switching Models

In this section, we analyze the theoretical performance of the regime-switching model when the true underlying regime sequences of history returns are known, i.e., the estimation of the regime using covariates (such as US CPI and GDP data) is accurate. For the sake of analysis, we need to make additional assumptions besides Assumption 1.

ASSUMPTION 2 (K-Regime). *For the underlying true distribution of interest in (5), $\mathbb{P}^* := \sum_{i \in [K]} w_k^* \mathbb{P}_k^*$ with $w_k^* > 0, \forall k \in [K]; \sum_{k \in [K]} w_k^* = 1$. $\{\mathbb{P}_k^*\}_{k \in [K]}$ are K different distributions.*

Here, $\{w_k^*\}_{k \in [K]}$ are different across time since the last regime j^* varies across time in Algorithm 2, and we have $w_k^* = a_{j^*,k}^*$, where $A^* = \{a_{j,k}^*\}_{j,k \in [K]}$ represents the underlying true transition probability matrix in Algorithm 2. At the t -th period in Step 2 - 5 in Algorithm 1, the underlying true distribution $\mathbb{P}^*$ in Assumption 3 refers to the distribution of the random return at the time $N + t$.

ASSUMPTION 3 (Light tail). *The underlying true distribution $\mathbb{P}^*$ is a light tail distribution, i.e., $\mathbb{E}^{\tilde{\mathbf{r}} \sim \mathbb{P}_k^*}[\exp(\|\tilde{\mathbf{r}}\|^a)] < \infty, \forall k \in [K]$ for some $a > 0$.*

The assumption above is necessary to guarantee the traditional concentration result for the empirical distribution in the literature (see Fournier and Guillin 2015, Esfahani and Kuhn 2018).

The main results are presented here.

THEOREM 2 (Coverage of True Distribution). *Under Assumptions 1, 2, 3, if the sample size $N \geq \frac{c_0^2}{\pi_* \gamma_{ps}} \log(\frac{2K}{\delta} \sqrt{\frac{1}{\pi_*}})$ (where γ_{ps} is its pseudo-spectral gap with the definition deferred in Appendix EC.2.1), then $\mathbb{P}^* \in \Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}(\boldsymbol{\theta})$ with a probability of at least $1 - \delta$ if the ball size*

$$\theta_k \geq \left(\frac{c_{k,1} \log(K/\delta) + c_{k,2}}{N_k} \right)^{\frac{1}{\bar{t}}} + \frac{c_0}{\min_{i \in [K], w_i \neq 0} w_i} \sqrt{\frac{K \log(2K/\delta)}{N \pi_*}} \max_{i \neq k} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*), \forall k \in [K], \quad (13)$$

where $c_{k,1}, c_{k,2}$ are constants only depending on $\mathbb{P}_k^*$, c_0 is a constant only depending on the Markov chain and $\pi_* = \min_{i \in [K]} \pi_i$.

We briefly discuss some of the key ideas in proving this result. We estimate $\{w_k^*\}_{k \in [K]}$ and $\{\mathbb{P}_k^*\}_{k \in [K]}$ by their empirical estimators, denoted by w_k and $\hat{\mathbb{P}}_k = \frac{1}{N_k} \sum_{n \in [N_k]} \delta_{r_{nk}}$, respectively. To ease the analysis, we first reconstruct $\mathbb{P}^*$ as a convex combination of K different new regimes $\hat{\mathbb{P}}_k^*$ with the coefficient of each $\hat{\mathbb{P}}_k^*$ being the estimated weight w_k . Specially, we show that $\mathbb{P}^* := \sum_{k \in [K]} w_k^* \mathbb{P}_k^*$ can be represented by $\sum_{k \in [K]} w_k \hat{\mathbb{P}}_k^*$, where $\hat{\mathbb{P}}_k^* = \sum_{i \in [K]} \alpha_{ik} \mathbb{P}_i^*$, $\forall k \in [K]$ with $\sum_{i \in [K]} \alpha_{ik} = 1$, $\alpha_{kk} = \min\{1, \frac{w_k^*}{w_k}\}$, $\forall k \in [K]$ and $\alpha_{ik} \geq 0$, $i, k \in [K]$. By doing so, we can build a connection between $\mathbb{P}^*$ and the empirical estimators (w_k and $\hat{\mathbb{P}}_k$), and reduce the analysis of the difference between $\mathbb{P}^* = \sum_{k \in [K]} w_k \hat{\mathbb{P}}_k^*$ and $\hat{\mathbb{P}} = \sum_{k \in [K]} w_k \hat{\mathbb{P}}_k$ to bounding the difference between $\hat{\mathbb{P}}_k^*$ and $\hat{\mathbb{P}}_k$ under each regime k , i.e., $W_m(\hat{\mathbb{P}}_k^*, \hat{\mathbb{P}}_k)$. By triangle inequalities $W_m(\hat{\mathbb{P}}_k^*, \hat{\mathbb{P}}_k) \leq W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k) + W_m(\hat{\mathbb{P}}_k^*, \mathbb{P}_k^*)$, we turn our attention to bounding $W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k)$ and $W_m(\hat{\mathbb{P}}_k^*, \mathbb{P}_k^*)$, respectively. The bound of $W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k)$ is a straightforward application of the measure concentration result in Esfahani and Kuhn (2018). This term characterizes the estimation error of $\mathbb{P}_k^*$ from its empirical distribution in each regime k . The second term ($W_m(\hat{\mathbb{P}}_k^*, \mathbb{P}_k^*)$) arises from the error in estimating the regime-switching behavior, i.e., $|w_k - w_k^*|, k \in [K]$ and its magnified effects on the difference of each $\mathbb{P}_i \neq \mathbb{P}_k$ from the Wasserstein distance, i.e., $W_m(\mathbb{P}_k^*, \mathbb{P}_i^*)$. Intuitively, when $|w_k - w_k^*| \rightarrow 0$, $W_m(\hat{\mathbb{P}}_k^*, \mathbb{P}_k^*) \rightarrow 0$. It leverages the estimation error rate of the transition matrix of the Markov chain in Wolfer and Kontorovich (2021) to bound this term.

REMARK 3. When $K = 1$, there is only one regime. We see the second term in the bound disappears. The result is consistent with the standard Wasserstein measure concentration results established in Esfahani and Kuhn (2018), which confirms the validity of our results to some extent.

REMARK 4. We claim that the bound in Theorem 2 cannot be further improved using other estimators rather than the empirical ones as we choose in this paper. Note that the bound is built on two key concentration results: the upper bound of the Wasserstein distance between the empirical distribution $\hat{\mathbb{P}}_k$ and the true distribution $\mathbb{P}_k^*$ under each regime k which is $O_p(N_k^{-\frac{1}{I}})$ (see Lemma EC.5), and the estimation error bound of the Markov chain's transition matrix $\sum_{k \in [K]} |w_k - w_k^*|$, which is $O_p(N^{-\frac{1}{2}})$ (see Lemma EC.3). These two bounds can be shown to be tight in the order of N . Specifically, according to Theorem 3.2 in Wolfer and Kontorovich (2021), for the set of Markov chains satisfying Assumption 1, 2, there exists one instance of Markov chain S such that for any estimator $\{\hat{w}_k\}_{k \in [K]}$ (as a function of history states $\{s_i\}_{i \in [N]}$), with probability at least $\frac{1}{10}$, $\sum_{k \in [K]} |\hat{w}_k -$

$w_k^*(S)$ is lower bounded by $C_1 \sqrt{\frac{K}{N\pi_k}}$ for some constant C_1 . Similarly, according to Singh and Póczos (2018), one can show that for the set of distributions $\mathcal{P}$ satisfying Assumption 3 and for any estimator $\hat{\mathbb{P}}$, we have

$$\inf_{\hat{\mathbb{P}}} \sup_{\mathbb{P}_k^* \in \mathcal{P}} \mathbb{E}_{\{\mathbf{r}_{nk}\}_{n \in [N_k]} \sim \mathbb{P}_k^*} [W_m(\mathbb{P}_k^*, \hat{\mathbb{P}})] \geq \frac{C_2}{N_k^{\frac{1}{I}}}, \quad (14)$$

for some constants C_2 .

REMARK 5. Note that when the sample size N is large enough, the first term will dominate the minimum required ball size in (13) since $\Theta((N_k)^{-\frac{1}{I}}) = \Theta((N\pi_k)^{-\frac{1}{I}}) > \Theta((N\pi_k)^{-\frac{1}{2}})$ since $\pi_k > 0$ from Assumption 1. Hence we use the first term which is in the order of $N^{-\frac{1}{I}}$ and choose a more conservative $\theta_k = \gamma N^{-\frac{1}{I}}$ with a tuning hyperparameter γ in our numerical studies.

Data-independent Measure Concentration. The lower bound of the choice of the ball size in (13) is a function of N_k, w_k , and thus depends on the observational regimes of history samples. To remove this dependence, we further take advantage of the concentration tools in bounding the empirical frequency of states in the Markov chain, i.e., $|\frac{N_k}{N} - \pi_k|$ (shown in Lemma EC.4 in Appendix EC.2) to revise the concentration guarantee. Specifically, we can get a minimum required ball size independent of N_k, w_k when the sample size N is large enough. Proposition 3 summarizes the results.

PROPOSITION 3 (**Data-independent Measure Concentration**). *Under Assumptions 1, 2, 3, $\forall \delta > 0$, and the Markov chain is reversible so that spectral gap γ_s (definition deferred in Appendix EC.2.1) is meaningful, if the sample size satisfies:*

$$N \geq \max \left\{ \frac{c_0^2}{\pi_* \gamma_{ps}} \log \left(\frac{2K}{\delta} \sqrt{\frac{1}{\pi_*}} \right), \frac{\log(4K/\delta)}{2c'_0 \pi_*^2}, \frac{c_0^2 K \log(2K/\delta)}{\pi_* (\min_{i \in [K]} w_i^*)^2} \right\}$$

then $\mathbb{P}^* \in \Pi_{\hat{\mathbf{r}}} \mathcal{U}_{rs}(\theta)$ with a probability of at least $1 - \delta$ if the ball size

$$\theta_k \geq \left(\frac{c_{k,1} \log(2K/\delta) + c_{k,2}}{2N\pi_k} \right)^{\frac{1}{I}} + \frac{c_0}{2 \min_{i \in [K]} w_i^*} \sqrt{\frac{K \log(4K/\delta)}{N\pi_*}} \max_{i \neq k} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*), \forall k \in [K], \quad (15)$$

where c'_0 is a positive constant and depends on the spectral gap λ_s .

Suppose the underlying true distribution is a mixture of $K = 2$ Gaussian distributions, i.e., $\mathbb{P}^* = w_1^* \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) + w_2^* \mathcal{N}(-\boldsymbol{\mu}, \boldsymbol{\Sigma})$. If the norm $m = 1$, from Theorem 2, the minimum

required ball size reduces to $\theta_k \geq \left(\frac{c_{k,1} \log(2/\delta) + c_{k,2}}{N_k} \right)^{\frac{1}{T}} + \frac{2c_0 \|\boldsymbol{\mu}\|_2^2}{\min\{w_1, w_2\}} \sqrt{\frac{2 \log(2/\delta)}{N \pi_*}}$, $k \in \{1, 2\}$ since $W_1(\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}), \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})) \leq \sqrt{\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|_2^2}$ by Dowson and Landau (1982). From the bound, we can see that if $\boldsymbol{\mu}$ is larger (implying the distribution in each regime k ($\mathbb{P}_k^*$) is very different from each other) or the frequency difference between w_1^*, w_2^* is larger (implying w_1^*, w_2^* significantly differs), then a larger ball size θ_k is required. On the other hand, if π_* is very small, which implies one regime happens with a very small probability, then we cannot obtain enough samples in this regime to accurately estimate its corresponding w_k . Therefore, it also requires a large ball size according to the bound.

Performance of the Obtained Portfolio. Now we are ready to analyze the performance of the obtained portfolio using our data-driven approach. Theorem 2 shows that when the ambiguity aversion level $\boldsymbol{\theta}$ is large enough, the ambiguity set will include the true distribution with a high probability. Based on it, we further show that under the same ambiguity aversion level, the obtained portfolio using our data-driven approach can generate the risk measured by CVaR no larger than a certain threshold from the true risk level. Specifically, we list our results below in Theorem 3.

THEOREM 3. *Denote the solution $\hat{\mathbf{x}}, \mathbf{x}^*$ to each optimization problem:*

$$\begin{aligned} \hat{\mathbf{x}} &\in \arg \min_{\mathbf{x} \in \mathcal{X}} \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} CVaR_{\eta}^{\mathbb{P}}(\mathbf{x}) \\ \mathbf{x}^* &\in \arg \min_{\mathbf{x} \in \mathcal{X}} CVaR_{\eta}^{\mathbb{P}^*}(\mathbf{x}). \end{aligned}$$

Then with probability $1 - \delta$, if θ_k satisfies (13), we would have:

$$CVaR_{\eta}^{\mathbb{P}^*}(\hat{\mathbf{x}}) - CVaR_{\eta}^{\mathbb{P}^*}(\mathbf{x}^*) \leq \frac{2\|\mathbf{x}^*\|_{m^*}}{1 - \eta} \sum_k w_k \theta_k,$$

where m means that we take ℓ_m -norm as the cost function in the Wasserstein distance.

In summary, Theorem 2 establishes the concentration result of the true distribution for the frequency-based approach, based on which, we can analyze the performance of the obtained portfolio using this frequency-based data-driven optimization approach (see Theorem 3). Note that in this analysis, we assume that the true regimes of history returns are observed. However, in practice, the true underlying regimes of history returns are unknown but inferred from algorithms. Therefore, we can no longer ensure the output state sequence is “almost” the same as the true regime sequence such that $W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k), \sum_{k \in [K]} |w_k - w_k^*| \rightarrow$

0 as $N \rightarrow \infty$. In this case, we use the HMM to dynamically determine the regime and estimate the transition probability matrix. Although HMM has shown its outstanding performance numerically, to the best of our knowledge, it is still an open problem to analyze its concentration performance theoretically. We leave it for future research.

5. Case Studies

In this section, we examine the performance of the proposed regime-switching distributionally robust CVaR portfolio using the datasets used in DeMiguel et al. (2009) with an extended test period and a dataset with different asset classes. All the data are monthly data. We compare our proposed portfolio with the benchmark portfolios: equally-weighted (EW) portfolio, which is a robust choice that is hard to beat (see DeMiguel et al. 2009), minimum variance portfolio, which outperforms classical mean-variance portfolio (see Jagannathan and Ma 2003), and the robust mean-CVaR portfolio with a moment-based ambiguity set where the first-two moments are specified (see Kang et al. 2019). We use the shorthand, “EW”, “MVP” and “DR Mean-CVaR (Moments)” for the aforementioned portfolios, respectively.

5.1. Datasets and Numerical Settings

We include the six datasets considered in DeMiguel et al. (2009). They are S&P sectors, Industry portfolios, International portfolios, Mkt/SMB/HML, FF 1-factor, and FF 4-factor; see DeMiguel et al. (2009) for the detailed descriptions of the datasets. The time periods of the datasets are up to November 2004 (from July 1963 at the earliest). We extend the datasets to April 2019 by downloading the same classes of data from Ken French’s website and MSCI (Morgan Stanley Capital International). Each asset’s returns can be directly input to Algorithm 2 (frequency-based approach). To get the input of Algorithm 3 (HMM), we need to generate observational sequences $o_t \in \{+, -\}$ based on the asset’s returns. Specifically, if the return is above the risk-free rate (given by the 90-day Treasury-bill yield) at time t , then $o_t = +$, otherwise, $o_t = -$. The empirical results for the six types of datasets over different periods (up to 2004 and up to 2019) are presented in Section 5.2 and Appendix EC.6.1, respectively.

In Section 5.3, we include another dataset comprising five different asset classes to explore the regime-switching property of the proposed portfolio. The five asset classes are (i) equities; (ii) 10-year treasury note; (iii) treasury inflation-protected securities

(TIPS); (iv) commodities; (v) credit. The asset prices are downloaded from various sources including BlackRock and Barclays Indices. The time period spans from January 2004 to April 2019. The class of equities is formed using the Global Industry Classification Standard (GICS), and the ten industries considered are Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Healthcare, Financials, Information-Technology, Telecommunications, and Utilities. Gold, Silver, and West Texas Intermediate (WTI) are selected to represent the commodity class. PIMCO Investment Grade Corporate Fund is selected to represent the credit class. All of them are selected mainly because they are liquid and tradable in the market. It will be interesting to see whether our portfolio can invest in the appropriate asset classes over a long time period. Besides using the assets' returns as input of Algorithm 2 (frequency-based approach), we also need to estimate regimes using either the bull and bear markets or the "weather" conditions by Bridgewater (2012) based on some macroeconomic variables (see construction details in Section 4.1.1) and input them to Algorithm 2. To get the input of Algorithm 3 (HMM), we need to generate new observational sequences $o_t \in \{1, 2, \dots, 5\}$ (see construction details in Section 4.1.2). Specifically, $o_t = i$ if the asset class i enjoys the highest return among all five assets at time t . In addi-

Shorthand of Strategy	Mean Constraint	Robustness	# of Regimes (K)	How the Regimes are determined
RSDR Mean-CVaR (HMM)	Yes	Yes	4	HMM with observations about five asset classes
RSDR Mean-CVaR (Weathers)	Yes	Yes	4	Frequency-based approach with Bridgewater's regimes
RSDR Mean-CVaR (Bull & Bear)	Yes	Yes	2	Frequency-based approach for bull and bear markets
RS Mean-CVaR (HMM)	Yes	No	4	HMM with observations about five asset classes
RS Mean-CVaR (Weathers)	Yes	No	4	Frequency-based approach with Bridgewater's regimes
RS Mean-CVaR (Bull & Bear)	Yes	No	2	Frequency-based approach for bull and bear markets
DR Mean-CVaR (Wasserstein)	Yes	Yes	1	N.A.
Mean-CVaR	Yes	No	1	N.A.
RSDR CVaR (HMM)	No	Yes	4	HMM with observations about five asset classes
RSDR CVaR (Weathers)	No	Yes	4	Frequency-based approach with Bridgewater's regimes
RSDR CVaR (Bull & Bear)	No	Yes	2	Frequency-based approach for bull and bear markets
RS CVaR (HMM)	No	No	4	HMM with observations about five asset classes
RS CVaR (Weathers)	No	No	4	Frequency-based approach with Bridgewater's regimes
RS CVaR (Bull & Bear)	No	No	2	Frequency-based approach for bull and bear markets
DR CVaR (Wasserstein)	No	Yes	1	N.A.
CVaR	No	No	1	N.A.

Table 1 CVaR Portfolios for Comparison.

tion to the aforementioned benchmarks, we also introduce a stochastic regime-switching model and a DRO portfolio optimization model with the traditional Wasserstein ambiguity set to examine the value of considering a robust model for the regime-switching model, and the value of adding the regime-switching feature in a DRO model, respectively. The shorthand and the corresponding specifications are presented in Table 1. We add a non-negative constraint, $x_i \geq 0$ for $i \in [I]$, into $\mathcal{X}$ for all the CVaR and Mean-CVaR portfolios.

The Mean-CVaR portfolios are obtained by adding first-moment constraint $\mu'x \geq R$ to the formulation (11), where μ is estimated by the sample mean vector and R is a target. We set R to the 40% lower quantile of the historical returns of all assets for the respective datasets. We choose the set of candidate ambiguity aversion level $\Theta = \{\theta \in \mathbb{R}^K : \theta_k = \gamma N^{-\frac{1}{T}}, \forall k \in [K]; \gamma \in \{0.02, 0.04, \dots, 0.1\}\}$ and adopt cross-validation shown in Appendix EC.4 to tune γ for all DRO models in our experiments. For the DeMiguel et al. (2009)'s datasets across different time periods, we compare only EW, RS CVaR (HMM), DR CVaR (Wasserstein), and RSDR CVaR (HMM) due to the space limit, while for our own dataset, we compare all the portfolios.

We measure the performance of each model using four commonly used performance metrics in the finance literature: the out-of-sample Sharpe ratio, certainty equivalent (CEQ) return, maximum drawdown, and turnover. We refer interested readers to Appendix EC.5 for their definitions. Generally, a good portfolio should have a high Sharpe ratio and CEQ and a low maximum drawdown and turnover.

5.2. Empirical Results of DeMiguel et al. (2009)'s Datasets During the Same Time

We compare the performance of our proposed portfolios with the same considerations of DeMiguel et al. (2009). We only report the 1/N strategy (EW) from their paper because it is the best amongst the rest in their paper. Moreover, EW portfolio is an extreme case of the (RS)DR CVaR portfolios as suggested in Proposition 2. Our empirical results are presented in Table 2.

We first evaluate the marginal benefits of regime-switching and DRO models by comparing our approach with the RS CVaR (HMM) and DR CVaR (Wasserstein) portfolios. Without robustness, RS CVaR (HMM) appears to be more sensitive than RSDR CVaR (HMM) since one can see that the performance of RS CVaR (HMM) is volatile compared with EW and RSDR CVaR (HMM) across different datasets. It demonstrates the efficiency of incorporating robustness into the regime-switching model to mitigate the errors from estimating non-stationary parameters. DR CVaR (Wasserstein) performs analogous to RSDR CVaR (HMM) in the tested data. We can even see an evident decrease in the turnover from DR CVaR (Wasserstein) to RSDR CVaR (HMM), which implies that adding robustness for non-stationary optimization models can stabilize the positions of portfolios. The good performance of DR CVaR (Wasserstein) is mainly due to the asset structure. Each data set in DeMiguel et al. (2009) is from only one class, and the return changes of each asset across the dataset are highly similar. Hence the non-stationarity behavior incorporating the regime behavior does not bring significant gains to the original model. In contrast, when there are

Strategy	S&P sectors $I = 11$	Industry sectors $I = 11$	Inter'l portfolios $I = 9$	Mkt/ SMB/HML $I = 3$	FF 1-factor $I = 21$	FF 4-factor $I = 24$
1. Sharpe Ratio						
EW	0.1876	0.1353	0.1277	0.2240	0.1623	0.1753
RS CVaR (HMM)	0.0811	0.1387	0.0947	0.2073	0.1716	0.2554
DR CVaR (Wasserstein)	0.1876	0.1368	0.1258	0.2240	0.1844	0.2334
RSDR CVaR (HMM)	0.1879	0.1369	0.1304	0.2240	0.1831	0.2185
2. CEQ						
EW	0.0069	0.0050	0.0046	0.0039	0.0073	0.0072
RS CVaR (HMM)	0.0025	0.0047	0.0032	0.0035	0.0069	0.0046
DR CVaR (Wasserstein)	0.0069	0.0051	0.0045	0.0039	0.0080	0.0055
RSDR CVaR (HMM)	0.0069	0.0050	0.0047	0.0039	0.0080	0.0063
3. Maximum Drawdown						
EW	0.3147	0.4488	0.3941	0.1732	0.4466	0.3896
RS CVaR (HMM)	0.3860	0.4234	0.3903	0.1912	0.3969	0.1302
DR CVaR (Wasserstein)	0.3473	0.4495	0.3966	0.1729	0.4172	0.2653
RSDR CVaR (HMM)	0.3056	0.4485	0.3958	0.1732	0.4228	0.2941
4. Turnover						
EW	0.0376	0.0244	0.0334	0.0266	0.0190	0.0226
RS CVaR (HMM)	0.4266	0.3481	0.4010	0.1010	0.5461	0.2081
DR CVaR (Wasserstein)	0.0376	0.0286	0.0381	0.0266	0.0556	0.1729
RSDR CVaR (HMM)	0.0790	0.0289	0.0510	0.0266	0.1253	0.1964

Table 2 Empirical Results of DeMiguel et al. (2009)'s datasets during the same time.

multiple assets in the data set in Section 5.3, RSDR CVaR (HMM) shows its significant performance improvement over DR CVaR (Wasserstein) since different asset classes can behave quite differently under different regimes characterized by Bridgewater (2012) (see Figure 1). In summary, RS CVaR (HMM) and DR CVaR (Wasserstein) perform quite differently in different data sets with no clear winner. RSDR CVaR (HMM) appears to be a robust choice combining the merits of those two.

It can also be seen that the RSDR CVaR (HMM) uniformly outperforms the EW portfolio in terms of Sharpe ratio over all datasets. It is worthwhile to note that DeMiguel et al. (2009) has shown that none of the model-based strategies, including those incorporating the Bayesian framework, portfolio constraints, and various combinations, can uniformly outperform EW portfolio. Therefore, it implies that our proposed RSDR CVaR (HMM) outperforms all the portfolios considered in DeMiguel et al. (2009) in terms of Sharpe ratio. However, the turnovers of the proposed CVaR models compared to the EW portfolio are generally higher. It is reasonable because the proposed models allocate portfolio weights via risk parity, and each asset class has a different risk based on the regime, which implies that there are more occasions to rebalance the portfolio for these models compared to the EW portfolio since the regime changes across time.

We further extend the studies to a dataset that is extended from DeMiguel et al. (2009)'s data to the first quarter of 2019, covering 2008 financial crisis. In general, we observe similar results. We defer the details to Appendix EC.6.1.

Strategy	Sharpe Ratio	CEQ	Max Drawdown	Turnover
RSDR Mean-CVaR (HMM)	0.1406	0.0036	0.3252	0.2452
RSDR Mean-CVaR (Weathers)	0.1197	0.0031	0.3463	0.1844
RSDR Mean-CVaR (Bull & Bear)	0.1293	0.0033	0.2931	0.1695
RS Mean-CVaR (HMM)	0.0818	0.0016	0.2714	0.5866
RS Mean-CVaR (Weathers)	0.0329	0.0005	0.2644	0.5712
RS Mean-CVaR (Bull & Bear)	0.0232	0.0003	0.2503	0.2043
DR Mean-CVaR (Wasserstein)	0.0788	0.0018	0.3369	0.0811
Mean-CVaR	0.0140	0.0001	0.2890	0.1854
RSDR CVaR (HMM)	0.1373	0.0035	0.3348	0.2490
RSDR CVaR (Weathers)	0.1177	0.0031	0.3540	0.1831
RSDR CVaR (Bull & Bear)	0.1272	0.0032	0.2930	0.1887
RS CVaR (HMM)	0.0770	0.0015	0.2714	0.5812
RS CVaR (Weathers)	0.0366	0.0006	0.2644	0.5614
RS CVaR (Bull & Bear)	0.0110	(0.0000)	0.2796	0.1934
DR CVaR (Wasserstein)	0.1081	0.0026	0.3197	0.1924
CVaR	0.0140	0.0001	0.2694	0.1854
EW	0.1090	0.0029	0.3830	0.0376
MVP	0.0297	0.0003	0.2020	0.0995
DR Mean-CVaR (Moments)	0.1185	0.0023	0.2890	0.0515

Table 3 Empirical Results for Tradable Assets of Different Classes from 2004 to 2019 Q1

5.3. Empirical Results for Tradable Assets of Different Classes from 2004 to 2019

Finally, we investigate whether the RSDR Mean-CVaR (HMM) is robust across different datasets, and how the regime-switching ambiguity model leads to a smart portfolio allocation under different market conditions. To this end, we propose a dataset with multiple asset classes constructed in Section 5.1. We also include other models for comparison and observe that other variations of the model are dominated by RSDR Mean-CVaR (HMM). Note that in this study, we take $M = 36$ since the earliest data we can get starts from January 2004, and we want to analyze our model during the 2008-2009 great recession period.

We present the empirical results for model comparison in Table 3. From the table, we first observe that the RSDR Mean-CVaR (HMM) performs the best among all the models in terms of Sharpe ratio and CEQ. Second, all regime-switching models based on the market and macroeconomic states outperform other benchmarks. Moreover, by comparing our RSDR CVaR models with DR CVaR models, one can see a significant performance improvement by taking regime-switching into account in the DR model. Third, we note that portfolios with RS or DR alone can hardly outperform any benchmark, but RSDR portfolios, especially with HMM, promise a huge advantage over all three benchmarks in terms of Sharpe ratio. It is also noteworthy that when we incorporate the distributionally robustness (see the comparison between RS and RSDR models) for each scenario determination, the turnover rate drops by a large margin while the out-of-sample portfolio return

and Sharpe ratio are often higher. Finally, our proposed HMM approach is generally better than the frequency-based approach.

To better illustrate our point of view, we further visualize the portfolio wealth changes over time with an initial wealth of unit 1. We compare two clusters of Mean-CVaR models and CVaR models using different approaches to determining the regimes. The results of our proposed HMM approaches are shown in Figure 2. The frequency-based approaches present an analogous pattern, and we report the results in Appendix EC.6.3. One can see that the proposed regime-switching ambiguity set outperforms the conventional ambiguity set. Here, the noticeable superiority of regime-switching models is ascribed to the multi-class feature of the dataset. Note that in such data, different asset classes such as equities, commodities, and credit may behave differently under different regimes characterized by Bridgewater (2012) (see Figure 1). The superior performance of our proposed approach in this multi-class dataset indicates that our method's advantage becomes more significant if the data contain multiple sources of time-varying uncertainties.

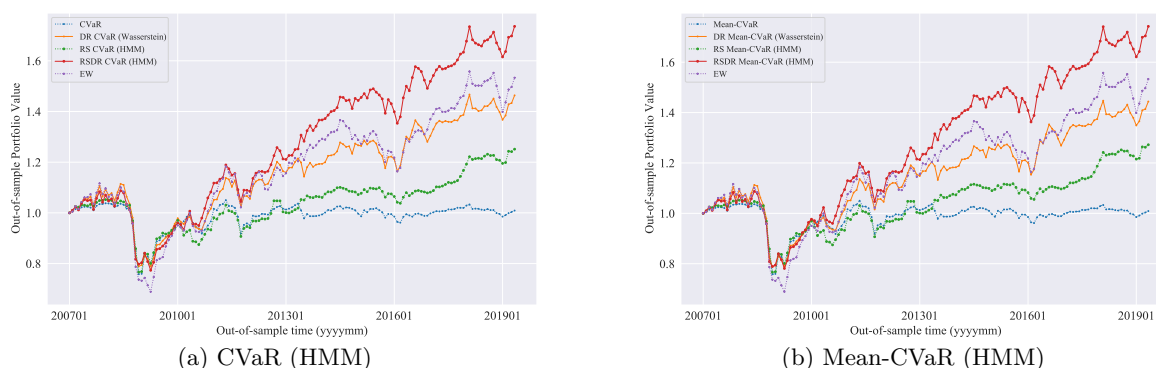


Figure 2 Comparison of different HMM models.

Finally, note that the financial crisis in 2008 brought an apparent change in the financial regime. It is interesting to investigate the portfolio's performance during 2008-2009 only. Figure 3 shows how the portfolio weights of different asset classes varied during 2008 for the RSDR CVaR (Weathers). According to the figure, our model can identify the regime change starting around November 2008 and adjust the portfolio weight such that the allocation to asset classes with lower risks, like gold and bond, rises over time and asset classes with higher risks like equities fall over time. (Note that during the recessionary environment, equities yield negative returns but gold and bond yield positive returns in Figure 1.) Therefore, this allows our model to outperform the EW portfolio and S&P 500 Index in terms of annual returns, which are -33.25% and -38.49%, respectively, while RSDR Mean-CVaR (HMM, Weathers, Bull & Bear) yields 'only' -25.85%, -30.03%, -25.14% annual return in 2008, respectively. Our modeling framework of the regime-switching ambiguity set helps

investors allocate their portfolio such that the weightage of highly risky asset classes falls over time to maximize risk-weighted returns in distressed years. We also test different parameter values under the framework of RSDR CVaR models. The obtained insights are similar. We refer interested readers to Appendix EC.6.4 for the detailed results.

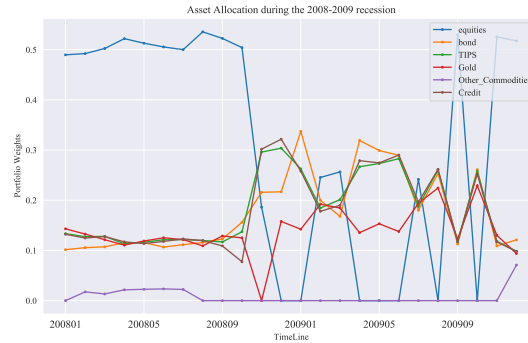


Figure 3 Asset allocation for RSDR CVaR (Weathers) during the 2008-2009 great recession ($\gamma = 0.01$).

In summary, from the performance comparison across different data, we can see that RSDR CVaR (HMM) is very robust in achieving good performance in various data sets. We believe that the superior performance of our algorithm can be observed under other datasets or parameter settings. This is because our intervention in portfolio construction is minimal. We only tune the hyperparameter γ (related to robustness) via cross-validation (see Appendix EC.4). From the performance comparison in single-class and multi-class data, we can see that the constructed ambiguity set is well-designed for non-stationary financial time series data. Its credibility is mainly linked to the persistence of multiple economic regimes and model uncertainty in each regime. It is one of the insights we bring to the literature.

6. Conclusions

We formulate a general framework to model the regime-switching nature of uncertainties in a decision-making environment from a distributionally robust perspective. In particular, we construct a regime-switching ambiguity set and propose two methods to estimate the transition probability and determine the portfolio allocation. We prove that the proposed DRO problem can be reformulated as a linear program (or second-order cone program) that can be efficiently solved. Moreover, we prove that our proposed ambiguity set can include the underlying distribution of random returns with high probability under mild assumptions. To further validate our method, we also numerically demonstrate that our proposed portfolios significantly outperform the benchmark portfolios across various datasets. The improvement is attributed to the regime-switching feature of the model to

allocate portfolio weightings based on the risk, which depends on the current environment. In addition, we examine the marginal contributions of regime-switching and DRO models, which validate the necessity of each of them as well as the benefits of merging them, as in our proposal. In summary, our proposed model, RSDR Mean-CVaR (HMM), helps investors allocate their portfolio to achieve a better Sharpe ratio and CEQ than other models.

This study can serve as the basis for future studies. First, as we have seen in the case studies that the turnover of the models is higher than that of the EW portfolio. Future research can consider adding the transaction cost in our formulation to potentially lower the turnover rate. Second, note that we consider either two or four regimes in the HMM in this paper. We can optimize the number of hidden regimes and use other classification methods to classify the regimes. Finally, we can consider extending the current framework to other operations management problems such as supply chain disruptions, where non-stationarity is a salient issue.

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References

- Bae, Geum Il, Woo Chang Kim, John M Mulvey. 2014. Dynamic asset allocation for varied financial markets under regime switching framework. *European Journal of Operational Research* **234**(2) 450–458.
- Ban, Gah-Yi, Nouredine El Karoui, Andrew E. B. Lim. 2018. Machine learning and portfolio optimization. *Management Science* **64**(3) 1136–1154. doi:10.1287/mnsc.2016.2644.
- Baum, Leonard E. 1972. An inequality and associated maximization technique in statistical estimation for probabilistic functions of Markov processes. *Inequalities III: Proceedings of the 3rd Symposium on Inequalities*, vol. 3. Academic Press, University of California, Los Angeles, 1–8.
- Baum, Leonard E., Ted Petrie. 1966. Statistical inference for probabilistic functions of finite state Markov chains. *The Annals of Mathematical Statistics* **37**(6) 1554–1563. doi:10.1214/aoms/1177699147.
- Bazzi, Marco, Francisco Blasques, Siem Jan Koopman, Andre Lucas. 2016. Time-varying transition probabilities for markov regime switching models. *Journal of Time Series Analysis* **38**(3) 458–478. doi:10.1111/jtsa.12211.
- Ben-Tal, Aharon, Dick den Hertog, Anja De Waegenaere, Bertrand Melenberg, Gijs Rennen. 2013. Robust solutions of optimization problems affected by uncertain probabilities. *Management Science* **59**(2) 341–357. doi:10.1287/mnsc.1120.1641.
- Bertsimas, Dimitris, Bart Van Parys. 2021. Bootstrap robust prescriptive analytics. *Mathematical Programming* 1–40.
- Blanchet, Jose, Lin Chen, Xun Yu Zhou. 2022. Distributionally robust mean-variance portfolio selection with Wasserstein distances. *Management Science* **68**(9) 6382–6410.
- Boyd, Stephen, Lieven Vandenbergh. 2004. *Convex optimization*. Cambridge University Press.
- Bridgewater. 2012. The all weather story. Tech. rep., Bridgewater Associates, LP. URL <https://www.bridgewater.com/research-library/the-all-weather-strategy/>.
- Chang, Yoosoon, Yongok Choi, Joon Y. Park. 2017. A new approach to model regime switching. *Journal of Econometrics* **196**(1) 127–143. doi:10.1016/j.jeconom.2016.09.005.

- Chen, Zhi, Melvyn Sim, Peng Xiong. 2020. Robust stochastic optimization made easy with RSOME. *Management Science* **66**(8) 3329–3339. doi:10.1287/mnsc.2020.3603.
- Costa, Giorgio, Roy Kwon. 2020. A regime-switching factor model for mean-variance optimization. *Journal of Risk* .
- Costa, Giorgio, Roy H Kwon. 2019. Risk parity portfolio optimization under a markov regime-switching framework. *Quantitative Finance* **19**(3) 453–471.
- Delage, Erick, Yinyu Ye. 2010. Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Operations Research* **58**(3) 595–612. doi:10.1287/opre.1090.0741.
- DeMiguel, Victor, Lorenzo Garlappi, Raman Uppal. 2009. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies* **22**(5) 1915–1953. doi:10.1093/rfs/hhm075.
- Dempster, A. P., N. M. Laird, D. B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)* **39**(1) 1–22. doi:10.1111/j.2517-6161.1977.tb01600.x.
- Dou, Xialiang, Mihai Anitescu. 2019. Distributionally robust optimization with correlated data from vector autoregressive processes. *Operations Research Letters* **47**(4) 294–299.
- Dowson, DC, BV666017 Landau. 1982. The fréchet distance between multivariate normal distributions. *Journal of multivariate analysis* **12**(3) 450–455.
- Esfahani, Peyman Mohajerin, Daniel Kuhn. 2018. Data-driven distributionally robust optimization using the Wasserstein metric: performance guarantees and tractable reformulations. *Mathematical Programming* **171**(1-2) 115–166. doi:10.1007/s10107-017-1172-1.
- Esteban-Pérez, Adrián, Juan M Morales. 2021. Distributionally robust stochastic programs with side information based on trimmings. *Mathematical Programming* 1–37.
- Fan, Jianqing, Bai Jiang, Qiang Sun. 2021. Hoeffding’s inequality for general markov chains and its applications to statistical learning. *J. Mach. Learn. Res.* **22** 139–1.
- Forney, G.D. 1973. The Viterbi algorithm. *Proceedings of the IEEE* **61**(3) 268–278. doi:10.1109/proc.1973.9030.
- Fouque, Jean-Pierre, Chi Seng Pun, Hoi Ying Wong. 2016. Portfolio optimization with ambiguous correlation and stochastic volatilities. *SIAM Journal on Control and Optimization* **54**(5) 2309–2338. doi:10.1137/15m1032533.
- Fournier, Nicolas, Arnaud Guillin. 2015. On the rate of convergence in wasserstein distance of the empirical measure. *Probability Theory and Related Fields* **162**(3) 707–738.
- Fu, Jun, Jiaqin Wei, Hailiang Yang. 2014. Portfolio optimization in a regime-switching market with derivatives. *European Journal of Operational Research* **233**(1) 184–192.
- Ghahramani, Zoubin. 2001. An introduction to hidden Markov models and Bayesian networks. *International Journal of Pattern Recognition and Artificial Intelligence* **15**(01) 9–42. doi:10.1142/s0218001401000836.
- Guidolin, Massimo, Allan Timmermann. 2007. Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control* **31**(11) 3503–3544.
- Hamilton, James D. 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* **57**(2) 357–384. doi:10.2307/1912559.
- Hamilton, James D. 2010. Regime switching models. *Macroeconometrics and Time Series Analysis*. The New Palgrave Economics Collection, Palgrave Macmillan, London, 202–209. doi:10.1057/9780230280830_23.
- Hanasusanto, Grani A., Daniel Kuhn, Stein W. Wallace, Steve Zymmler. 2015. Distributionally robust multi-item newsvendor problems with multimodal demand distributions. *Mathematical Programming* **152**(1-2) 1–32. doi:10.1007/s10107-014-0776-y.
- Hansen, Lars Peter, Thomas J. Sargent, Gauhar Turmuhambetova, Noah Williams. 2006. Robust control and model misspecification. *Journal of Economic Theory* **128**(1) 45–90. doi:10.1016/j.jet.2004.12.006.

- Jagannathan, Ravi, Tongshu Ma. 2003. Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance* **58**(4) 1651–1683. doi:10.1111/1540-6261.00580.
- Jurafsky, Daniel, James H. Martin. 2008. *Speech and Language Processing*. 2nd ed. Prentice Hall Series in Artificial Intelligence, Pearson Education (US), Upper Saddle River, NJ, United States. URL https://www.ebook.de/de/product/7295607/daniel_jurafsky_james_h_martin_speech_and_language_processing.html.
- Kang, Zhilin, Xun Li, Zhongfei Li, Shushang Zhu. 2019. Data-driven robust mean-CVaR portfolio selection under distribution ambiguity. *Quantitative Finance* **19**(1) 105–121. doi:10.1080/14697688.2018.1466057.
- Krokhmal, Pavlo, Tanislav Uryasev, Jonas Palmquist. 2001. Portfolio optimization with conditional value-at-risk objective and constraints. *The Journal of Risk* **4**(2) 43–68. doi:10.21314/jor.2002.057.
- Li, Mengmeng, Tobias Sutter, Daniel Kuhn. 2021. Distributionally robust optimization with markovian data. *arXiv preprint arXiv:2106.06741*.
- Litterman, Bob. 2003. *Modern Investment Management*. John Wiley & Sons. URL https://www.ebook.de/de/product/2097546/bob_litterman_modern_investment_management.html.
- Liu, Jia, Zhiping Chen, Abdel Lisser, Zhuojia Xu. 2019. Closed-form optimal portfolios of distributionally robust mean-cvar problems with unknown mean and variance. *Applied Mathematics & Optimization* **79**(3) 671–693.
- Long, Daniel Zhuoyu, Jin Qi, Aiqi Zhang. 2021. Supermodularity in two-stage distributionally robust optimization.
- Luenberger, David G, Yinyu Ye, et al. 1984. *Linear and nonlinear programming*, vol. 2. Springer.
- Markowitz, Harry. 1952. Portfolio selection. *The Journal of Finance* **7**(1) 77–91. doi:10.1111/j.1540-6261.1952.tb01525.x.
- Michaud, Richard O. 1989. The Markowitz optimization enigma: Is ‘optimized’ optimal? *Financial Analysts Journal* **45**(1) 31–42. doi:10.2469/faj.v45.n1.31.
- Nystrup, Peter, Henrik Madsen, Erik Lindström. 2017. Long memory of financial time series and hidden markov models with time-varying parameters. *Journal of Forecasting* **36**(8) 989–1002.
- Oprisor, Razvan, Roy Kwon. 2021. Multi-period portfolio optimization with investor views under regime switching. *Journal of Risk and Financial Management* **14**(1) 3.
- Pflug, Georg Ch, Alois Pichler. 2014. *Multistage stochastic optimization*. Springer.
- Pflug, Georg Ch, Alois Pichler, David Wozabal. 2012. The 1/n investment strategy is optimal under high model ambiguity. *Journal of Banking & Finance* **36**(2) 410–417.
- Pun, Chi Seng. 2021. G-expected utility maximization with ambiguous equicorrelation. *Quantitative Finance* **21**(3) 403–419. doi:10.1080/14697688.2020.1777321.
- Rahimian, Hamed, Sanjay Mehrotra. 2019. Distributionally robust optimization: A review. *arXiv preprint arXiv:1908.05659*.
- Ramani, Sivaramakrishnan, Archis Ghate. 2022. Robust markov decision processes with data-driven, distance-based ambiguity sets. *SIAM Journal on Optimization* **32**(2) 989–1017.
- Rockafellar, R.Tyrrell, Stanislav Uryasev. 2002. Conditional Value-at-Risk for general loss distributions. *Journal of Banking & Finance* **26**(7) 1443–1471. doi:10.1016/s0378-4266(02)00271-6.
- Singh, Shashank, Barnabás Póczos. 2018. Minimax distribution estimation in wasserstein distance. *arXiv preprint arXiv:1802.08855*.
- Stratonovich, R. L. 1960. Conditional Markov processes. *Theory of Probability & Its Applications* **5**(2) 156–178. doi:10.1137/1105015.
- Sutter, Tobias, Bart PG Van Parys, Daniel Kuhn. 2020. A general framework for optimal data-driven optimization. *arXiv preprint arXiv:2010.06606*.
- Tu, Jun. 2010. Is regime switching in stock returns important in portfolio decisions? *Management Science* **56**(7) 1198–1215. doi:10.1287/mnsc.1100.1181.

- Viterbi, Andrew J. 1967. Error bounds for convolutional codes and an asymptotically optimum decoding algorithm. *IEEE Transactions on Information Theory* **13**(2) 260–269. doi:10.1109/tit.1967.1054010.
- Wiesemann, Wolfram, Daniel Kuhn, Melvyn Sim. 2014. Distributionally robust convex optimization. *Operations Research* **62**(6) 1358–1376. doi:10.1287/opre.2014.1314.
- Wolfer, Geoffrey, Aryeh Kontorovich. 2021. Statistical estimation of ergodic markov chain kernel over discrete state space. *Bernoulli* **27**(1) 532–553.
- Yang, Insoon. 2017. A convex optimization approach to distributionally robust markov decision processes with wasserstein distance. *IEEE control systems letters* **1**(1) 164–169.
- Yao, Haixiang, Zhongfei Li, Yongzeng Lai. 2013. Mean-CVaR portfolio selection: A nonparametric estimation framework. *Computers & Operations Research* **40**(4) 1014–1022. doi:10.1016/j.cor.2012.11.007.
- Zhou, Xiang, Yi Xiong, Ningyuan Chen, Xuefeng Gao. 2020. Regime switching bandits. *arXiv preprint arXiv:2001.09390* .
- Zhu, Shushang, Masao Fukushima. 2009. Worst-case conditional Value-at-Risk with application to robust portfolio management. *Operations Research* **57**(5) 1155–1168. doi:10.1287/opre.1080.0684.

Online Appendix to “Data-Driven Distributionally Robust CVaR Portfolio Optimization under Regime-Switching Ambiguity Set”

EC.1. Proof in Section 3

We first make the following notation. For any given $(\tilde{\mathbf{r}}, \tilde{\mathbf{u}}, \tilde{n}, \tilde{k}) \sim \mathbb{P}^+ \in \mathcal{U}_{rs}(\boldsymbol{\theta})$, we use $\Pi_{\tilde{\mathbf{r}}} \mathbb{P}^+$ to represent the distribution of that $\tilde{\mathbf{r}}$. And we use $\Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}(\boldsymbol{\theta})$ to represent the ambiguity set of $\tilde{\mathbf{r}}$ where $(\tilde{\mathbf{r}}, \tilde{\mathbf{u}}, \tilde{n}, \tilde{k}) \in \mathcal{U}_{rs}(\boldsymbol{\theta})$ for some $(\tilde{\mathbf{u}}, \tilde{n}, \tilde{k})$.

Proof of Proposition 1. We first consider our ambiguity set under each regime $k \in [K]$ below with each ball center $\hat{\mathbb{P}}_k$.

$$\mathcal{U}_{rs}^k(\theta_k) = \left\{ \mathbb{P} \in \mathcal{P}(\mathbb{R}^I \times \mathbb{R} \times [N_k]) \left| \begin{array}{l} (\tilde{\mathbf{r}}, \tilde{\mathbf{u}}, \tilde{n}) \sim \mathbb{P} \\ \mathbb{E}^{\mathbb{P}}[\tilde{\mathbf{u}}] \leq \theta_k \\ \mathbb{P}[\tilde{\mathbf{r}} \in \mathbb{R}^I, \rho(\tilde{\mathbf{r}}, \mathbf{r}_{nk}) \leq \tilde{\mathbf{u}} \mid n \in [N_k]] = 1 \\ \mathbb{P}[\tilde{n} = n] = \frac{1}{N_k}, \forall n \in [N_k] \end{array} \right. \right\} \quad (\text{EC.1})$$

First we show that $\Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}(\boldsymbol{\theta}) \subseteq \mathcal{U}_w(\sum_{k \in [K]} w_k \theta_k)$. That is to say, fix any $\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})$ and we write its projection over $\tilde{\mathbf{r}}$ as $\Pi_{\tilde{\mathbf{r}}} \mathbb{P} = \sum_{k \in [K]} w_k \mathbb{P}_k = \sum_{k \in [K]} w_k \sum_{n \in [N_k]} \frac{1}{N_k} \mathbb{P}_{nk}$, where $\mathbb{P}_k \in \Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}^k(\theta_k)$ is the conditional distribution of $\tilde{\mathbf{r}}$ under regime k and $\mathbb{P}_{nk}$ is the conditional distribution of $\tilde{\mathbf{r}}$ given the outcome of random scenario is n under that regime. We can then construct a joint distribution by the natural coupling $\nu = \sum_{k \in [K]} \frac{w_k}{N_k} \sum_{n \in [N_k]} \mathbb{P}_{nk} \otimes \delta_{\mathbf{r}_{nk}}$. Recall the definition of Wasserstein distance, we have:

$$W_m(\Pi_{\tilde{\mathbf{r}}} \mathbb{P}, \hat{\mathbb{P}}) \leq \mathbb{E}^{\nu}[\rho(\tilde{\mathbf{r}}, \mathbf{r})] = \sum_{k \in [K]} w_k \sum_{n \in [N_k]} \frac{1}{N_k} \mathbb{E}_{\tilde{\mathbf{r}} \sim \mathbb{P}_{nk}}[\rho(\tilde{\mathbf{r}}, \mathbf{r}_{nk})] \leq \sum_{k \in [K]} w_k \theta_k.$$

To show the other side, we argue by contradiction. We can construct one $\mathbb{P}$ such that $W_m(\mathbb{P}, \hat{\mathbb{P}}) = \sum_k w_k \theta_k$ but for that $\tilde{\mathbf{r}} \sim \mathbb{P}$, there is no associated $(\tilde{\mathbf{u}}, \tilde{n}, \tilde{k})$ such that $(\tilde{\mathbf{r}}, \tilde{\mathbf{u}}, \tilde{n}, \tilde{k}) \in \mathcal{U}_{rs}(\boldsymbol{\theta})$. Below is a simple counterexample when $I = 1$.

When $I = 1$, computing Wasserstein distance with any $m \in [1, +\infty]$ would lead to the same result. For simplicity, we take the case when $m = 1, K = 2$ and $w_k = \frac{1}{2}, N_k = 1 \ \forall k \in [K]$. We let $\hat{\mathbb{P}} = \frac{1}{2} \delta_{\mathbf{r}_{11}} + \frac{1}{2} \delta_{\mathbf{r}_{12}}$ and $\mathbb{P}^+ = \frac{1}{2} \delta_{\tilde{\mathbf{r}}_{11}} + \frac{1}{2} \delta_{\tilde{\mathbf{r}}_{12}}$, where we choose θ_1, θ_2 such that $\tilde{\mathbf{r}}_{11} = \mathbf{r}_{11} - (\theta_1 + \frac{\theta_2}{2})$ and $\tilde{\mathbf{r}}_{12} = \mathbf{r}_{12} + \frac{\theta_2}{2}$ with $\tilde{\mathbf{r}}_{11} < \mathbf{r}_{11} < \mathbf{r}_{11} + \theta_1 < \mathbf{r}_{12} < \tilde{\mathbf{r}}_{12}$. By moving the mass of $\tilde{\mathbf{r}}_{11}$ to $\mathbf{r}_{11}$ and $\tilde{\mathbf{r}}_{12}$ to $\mathbf{r}_{12}$, the total transportation costs are $\frac{1}{2} \rho(\tilde{\mathbf{r}}_{11}, \mathbf{r}_{11}) + \frac{1}{2} \rho(\tilde{\mathbf{r}}_{12}, \mathbf{r}_{12}) = \frac{\theta_1 + \theta_2}{2}$. Therefore, $W_m(\mathbb{P}^+, \hat{\mathbb{P}}) \leq \frac{\theta_1 + \theta_2}{2}$. However, $\mathbb{P}[\rho(\tilde{\mathbf{r}}, \mathbf{r}_{11}) \leq \theta_1] = 0, \forall \tilde{\mathbf{r}} \in \mathbb{P}^+$, which implies $\mathbb{P}^+ \not\subseteq \Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}(\boldsymbol{\theta})$. Therefore, we show that $\Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}(\boldsymbol{\theta})$ is a true subset of $\mathcal{U}_w(\sum_{k \in [K]} w_k \theta_k)$. $\square$

As a by-product of this proposition, we also obtain a simple ambiguity set representation of $\Pi_{\tilde{\mathbf{r}}}(\boldsymbol{\theta})$ from Wasserstein distance. This result is a regime-based extension of Theorem 2 in Chen et al. (2020). We would use this to prove the measure concentration result later.

LEMMA EC.1. *For all $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K)$, we have:*

$$\Pi_{\tilde{\mathbf{r}}} \mathcal{U}_{rs}(\boldsymbol{\theta}) = \left\{ \mathbb{P} : \mathbb{P} = \sum_{k \in [K]} w_k \mathbb{P}_k, W_m(\mathbb{P}_k, \hat{\mathbb{P}}_k) \leq \theta_k \right\}, \quad (\text{EC.2})$$

where $\hat{\mathbb{P}}_k = \frac{1}{N_k} \sum_{n \in [N_k]} \delta_{\mathbf{r}_{nk}}$.

Proof of Lemma EC.1. In fact by definition of $\mathcal{U}_{rs}^k(\theta_k)$ and $\mathcal{U}_{rs}(\boldsymbol{\theta})$, we have:

$$\begin{aligned}\Pi_{\bar{\mathbf{r}}}\mathcal{U}_{rs}(\boldsymbol{\theta}) &= \left\{ \mathbb{P} | \mathbb{P} := \sum_{k \in [K]} w_k \mathbb{P}_k, \mathbb{P}_k \in \Pi_{\bar{\mathbf{r}}}\mathcal{U}_{rs}^k(\theta_k), \forall k \in [K] \right\} \\ &= \left\{ \mathbb{P} | \mathbb{P} := \sum_{k \in [K]} w_k \mathbb{P}_k, W_m(\mathbb{P}_k, \hat{\mathbb{P}}_k) \leq \theta_k, \forall k \in [K] \right\},\end{aligned}$$

where the second equality follows by $\Pi_{\bar{\mathbf{r}}}\mathcal{U}_{rs}^k(\theta_k) = \{\mathbb{P}_k | \mathbf{r} \sim \mathbb{P}_k, W_m(\mathbb{P}_k, \hat{\mathbb{P}}_k) \leq \theta_k\}$ from Theorem 2 in Chen et al. (2020). $\square$

Proof of Theorem 1. We first consider the same problem:

$$\begin{aligned}\sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \mathbb{E}^{\mathbb{P}}[G(\mathbf{x}, \mathbf{r})] &= \sup_{(\mathbf{r}, u_k) \sim \tilde{\mathbb{P}}_{nk}} \sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \mathbb{E}^{\tilde{\mathbb{P}}_{nk}}[G(\mathbf{x}, \mathbf{r})] \\ &\quad \text{s.t. } \sum_{n \in [N_k]} \frac{w_k}{N_k} \mathbb{E}^{\tilde{\mathbb{P}}_{nk}}[u_k] \leq w_k \theta_k, \forall k \in [K], \\ &\quad \tilde{\mathbb{P}}_{nk}[\mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k] = 1, \forall n \in [N_k], k \in [K].\end{aligned}$$

Denote the dual variables of each constraint as $\beta_k \forall k \in [K]$ and $\hat{\alpha}_{nk} \forall n \in [N_k], k \in [K]$, respectively. We derive the dual problem under the proposed regime-switching ambiguity set as follows:

$$\begin{aligned}&\inf_{\hat{\alpha}, \beta} \sum_{k \in [K]} \sum_{n \in [N_k]} \hat{\alpha}_{nk} + \sum_{k \in [K]} \beta_k w_k \theta_k, \\ &\text{s.t. } \hat{\alpha}_{nk} + \frac{w_k}{N_k} u_k \beta_k \geq \frac{w_k}{N_k} G(\mathbf{x}, \mathbf{r}), \forall k \in [K], n \in [N_k], \mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k, \\ &\quad \beta_k \geq 0, \forall k \in [K]. \\ &= \inf_{\alpha, \beta} \sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \alpha_{nk} + \sum_{k \in [K]} \beta_k w_k \theta_k \\ &\quad \text{s.t. } \alpha_{nk} + u_k \beta_k \geq G(\mathbf{x}, \mathbf{r}), \forall k \in [K], n \in [N_k], \mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k, \\ &\quad \beta_k \geq 0, \forall k \in [K],\end{aligned}$$

where $\alpha_{nk} = \frac{N_k}{w_k} \hat{\alpha}_{nk}, \forall n \in [N_k], k \in [K]$. By the definition of $G(\mathbf{x}, \mathbf{r})$, we have

$$\begin{aligned}&\inf_{\alpha, \beta} \sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \alpha_{nk} + \sum_{k \in [K]} \beta_k w_k \theta_k \\ &\text{s.t. } \alpha_{nk} + \beta_k u_k \geq -\mathbf{x}' \mathbf{r} - v, \forall k \in [K], n \in [N_k], \mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k, \\ &\quad \alpha_{nk} + \beta_k u_k \geq 0, \forall k \in [K], n \in [N_k], \mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k, \\ &\quad \beta_k \geq 0, \forall k \in [K].\end{aligned} \tag{EC.3}$$

There are infinite constraints in this problem. To get a tractable reformulation, we take the first set of infinity constraints as an example to illustrate how to reformulate it into finite constraints.

The constraint is equivalent to $Z_1^{nk} := \inf_{\mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k} -\mathbf{x}' \mathbf{r} + \beta_k u_k \geq -\alpha_{nk} - v, \forall n \in [N_k], k \in [K]$,

where Z_1^{nk} can be reformulated as a set of linear constraints. We first show a technical lemma in Boyd and Vandenberghe (2004) for conic optimization.

LEMMA EC.2 (Example 5.12 in Boyd and Vandenberghe (2004)). *We consider the following conic optimization problem,*

$$\min_{\mathbf{p}_{nk}, u_k} -\mathbf{x}' \mathbf{p}_{nk} + \beta_k u_k - \mathbf{r}'_{nk} \mathbf{x}, \text{ s.t. } \mathbf{G} \begin{pmatrix} \mathbf{p}_{nk} \\ u_k \end{pmatrix} = \mathbf{f}, \begin{pmatrix} \mathbf{p}_{nk} \\ u_k \end{pmatrix} \in \bar{K}, \tag{EC.4}$$

where $\bar{K} \subseteq \mathbb{R}^{I+1}$ is a proper cone and $\mathbf{G} \in \mathbb{R}^{t \times n}, \mathbf{f} \in \mathbb{R}^t$ for some t . Its duality problem is as follows:

$$\max_{\boldsymbol{\gamma} \in \mathbb{R}^t} \mathbf{f}' \boldsymbol{\gamma} - \mathbf{r}'_{nk} \mathbf{x}, \text{ s.t. } \begin{pmatrix} -\mathbf{x} \\ \beta_k \end{pmatrix} - \mathbf{G}' \boldsymbol{\gamma} \in \bar{K}^*, \tag{EC.5}$$

where $\bar{K}^* = \{\mathbf{y} \mid \mathbf{y}'\mathbf{x} \geq 0, \forall \mathbf{x} \in \bar{K}\}$ denote the dual cone of $\bar{K}$. Furthermore, if there exists $\begin{pmatrix} \mathbf{p}_{nk} \\ u_k \end{pmatrix} \succ_{\bar{K}} \mathbf{0}$ with $\mathbf{G} \begin{pmatrix} \mathbf{p}_{nk} \\ u_k \end{pmatrix} = \mathbf{f}$, then strong duality holds.

If $u_k = 0$, Z_1^{nk} reduces to $-\mathbf{r}'_{nk}\mathbf{b}_{nk} \geq -\alpha_{nk} - v$. Otherwise, we introduce the auxiliary variable $\mathbf{p}_{nk} = \mathbf{r} - \mathbf{r}_{nk}$. And the left-hand side of Z_1^{nk} can be reformulated as:

$$\min_{\mathbf{p}_{nk}, u_k} -\mathbf{x}'\mathbf{p}_{nk} + \beta_k u_k - \mathbf{r}'_{nk}\mathbf{x}, \text{ s.t. } \|\mathbf{p}_{nk}\|_m \leq u_k. \quad (\text{EC.6})$$

When we consider $\bar{K} = \left\{ \begin{pmatrix} \mathbf{p}_{nk} \\ u_k \end{pmatrix} \in \mathbb{R}^{I+1} \mid \|\mathbf{p}_{nk}\|_m \leq u_k \right\}$. (EC.6) is a special case of the conic optimization problem in Lemma EC.2 as $\begin{pmatrix} \mathbf{p}_{nk} \\ u_k \end{pmatrix} \in \bar{K}$. For $\begin{pmatrix} \mathbf{p}_{nk} \\ u_k \end{pmatrix} \in \bar{K}$, we have $\begin{pmatrix} \frac{\mathbf{p}_{nk}}{2} \\ \frac{u_k}{2} \end{pmatrix} \succ_{\bar{K}} \mathbf{0}$. Then strong duality holds. It is clear to show $\bar{K}^* = \left\{ \begin{pmatrix} \mathbf{x} \\ \beta_k \end{pmatrix} \in \mathbb{R}^{I+1} \mid \|\mathbf{x}\|_{m^*} \leq \beta_k \right\}$.

Using Lemma EC.2, $Z_1^{nk} \geq -\alpha_{nk} - v$ can be reformulated as the following constraints:

$$\begin{aligned} -\mathbf{r}'_{nk}\mathbf{x} &\geq -\alpha_{nk} - v, \forall n \in [N_k], k \in [K], \\ \beta_k &\geq \|\mathbf{x}\|_{m^*}, \forall k \in [K], \end{aligned}$$

Following a similar approach, we can convert the remaining constraints into a sequence of finite constraints. $Z_2^{nk} := \inf_{\mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k} \alpha_{nk} + \beta_k u_k \geq 0$ can be reformulated as:

$$\alpha_{nk} \geq 0, \forall n \in [N_k], k \in [K].$$

Combining all the reformulated constraint in (EC.3), we obtain the following LP:

$$\begin{aligned} \min_{v, \mathbf{x}, \alpha, \beta} \quad & v + \frac{1}{(1-\eta)} \left[\sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \alpha_{nk} + \sum_{k \in [K]} \beta_k w_k \theta_k \right] \\ \text{s.t.} \quad & \mathbf{r}'_{nk}\mathbf{x} \geq -\alpha_{nk} - v, \forall n \in [N_k], k \in [K], \\ & \beta_k \geq \|\mathbf{x}\|_{m^*}, \forall k \in [K], \\ & \alpha_{nk} \geq 0, \forall n \in [N_k], k \in [K], \\ & \mathbf{x} \in \mathcal{X}. \end{aligned} \quad (\text{EC.7})$$

Note that in (EC.7), if there exists $\beta_i^* \neq \beta_j^*$ for some $i, j \in [K]$ in the optimal solution, then we can always reduce both $\beta_i, \beta_j := \min(\beta_i^*, \beta_j^*)$ and keeping other variables the same to obtain a new feasible solution while reducing its objective value since $w_k \theta_k > 0, \forall k \in [K]$. Therefore, we have an implicit constraint $\beta_k = \beta, \forall k \in [K]$. Combining it with the current constraint, we obtain the reformulation, i.e. (12).

Then we show its computational tractability in some cases (i.e. $m = 1$ and $m = 2$). When $m = 1$, the second set of constraints in (12) is equivalent to:

$$\beta \geq x_i, \forall i \in [I],$$

$$\beta \geq -x_i, \forall i \in [I].$$

When $m = 2$, the second set of constraints in (12) is equivalent to:

$$(\beta, \mathbf{x}) \in \Omega_{soc}. \quad \square$$

Proof of Proposition EC.1. We have observed that the dual formula of the objective function can be eventually formulated as a linear program. We apply the same technique to handle the target constraint for the worst-case expected return. Consider

$$\begin{aligned} \inf_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \mathbb{E}^{\mathbb{P}}[\mathbf{r}'\mathbf{x}] &= \inf_{(\mathbf{r}, u_k) \sim \tilde{\mathbb{P}}_{nk}} \sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \mathbb{E}^{\tilde{\mathbb{P}}_{nk}}[\mathbf{r}'\mathbf{x}] \\ \text{s.t.} \quad & \sum_{n \in [N_k]} \frac{w_k}{N_k} \mathbb{E}^{\tilde{\mathbb{P}}_{nk}}[u_k] \leq w_k \theta_k, \forall k \in [K], \\ & \tilde{\mathbb{P}}_{nk}[\mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k] = 1, \forall n \in [N_k], k \in [K]. \end{aligned}$$

Denote the dual variables of each constraint as $\nu_k \forall k \in [K]$ and $\hat{\eta}_{nk} \forall n \in [N_k], k \in [K]$ respectively. We derive the dual problem under the proposed regime-switching ambiguity set as follow:

$$\begin{aligned}
& \sup_{\nu, \hat{\eta}} \sum_{k \in [K]} \sum_{n \in [N_k]} \hat{\eta}_{nk} - \sum_{k \in [K]} \nu_k w_k \theta_k \\
& \text{s.t. } \hat{\eta}_{nk} - \frac{w_k}{N_k} u_k \nu_k \leq \frac{w_k}{N_k} \mathbf{r}' \mathbf{x}, \forall n \in [N_k], k \in [K], \mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k, \\
& \quad \nu_k \geq 0, \forall k \in [K] \\
& = \sup_{\nu, \eta} \sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \eta_{nk} - \sum_{k \in [K]} \nu_k w_k \theta_k \\
& \text{s.t. } \eta_{nk} - u_k \nu_k \leq \mathbf{r}' \mathbf{x}, \forall n \in [N_k], k \in [K], \mathbf{r} \in \mathbb{R}^I, \rho(\mathbf{r}, \mathbf{r}_{nk}) \leq u_k, \\
& \quad \nu_k \geq 0, \forall k \in [K].
\end{aligned}$$

where $\eta_{nk} = \frac{N_k}{w_k} \hat{\eta}_{nk}$. There are infinite constraints. Following a similar analysis in Theorem 1, for $m \in [1, +\infty]$,

$$\begin{aligned}
\inf_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \mathbb{E}^{\mathbb{P}}[\mathbf{r}' \mathbf{x}] &= \sup_{\nu, \eta} \sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \eta_{nk} - \nu \sum_{k \in [K]} w_k \theta_k \\
&\text{s.t. } \mathbf{r}'_{nk} \mathbf{x} \geq \eta_{nk}, \forall n \in [N_k], k \in [K], \\
&\quad \nu \geq \|\mathbf{x}\|_{m^*}, \\
&\quad \nu \geq 0.
\end{aligned}$$

We can change the second constraint above similarly to reformulate as a set of linear and second order cone constraint(s) when $m = 1$ and $m = 2$ respectively. $\square$

Proof of Proposition 2. The proof is similar to what has been shown in Pflug et al. (2012), where they prove the case when $K = 1$. Denote the center of Wasserstein ambiguity set as $\hat{\mathbb{P}} \in \mathcal{U}_{rs}(\mathbf{0})$. We rewrite the objective of our function (11) to be $\min_{\mathbf{x} \in \mathcal{X}} \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_{\eta}^{\mathbb{P}}(\mathbf{x})$ as CVaR metric implicitly minimize $v \in \mathbb{R}$. Given the final objective of Theorem 1, for any fixed portfolio weight $\mathbf{x}^0$, we have:

$$\sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_{\eta}^{\mathbb{P}}(\mathbf{x}^0) = \text{CVaR}_{\eta}^{\hat{\mathbb{P}}}(\mathbf{x}^0) + \frac{1}{1 - \eta} \|\mathbf{x}^0\|_{m^*} \sum_{k \in [K]} w_k \theta_k, \quad (\text{EC.8})$$

where m^* is the conjugate norm of our proposed ℓ_m -norm used in the Wasserstein distance.

Equation (EC.8) can be obtained by letting the set of portfolio weight to have only one element, i.e., $\mathcal{X} = \{\mathbf{x}^0\}$ in Theorem 1. Then $\beta^* = \|\mathbf{x}^0\|_{m^*}$ and $\alpha_{nk}^* = (-\mathbf{r}'_{nk} \mathbf{x}^0 - v)^+$ in the optimal solution of (12) and by definition of CVaR under $\hat{\mathbb{P}}$, we have:

$$\text{CVaR}_{\eta}^{\hat{\mathbb{P}}}(\mathbf{x}^0) = v + \frac{1}{1 - \eta} \sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} (-\mathbf{r}'_{nk} \mathbf{x}^0 - v)^+,$$

which matches the right-hand side of (EC.8).

Then for the EW portfolio $\mathbf{x}^u = (\frac{1}{I}, \dots, \frac{1}{I})'$, we would obtain a sufficient condition for the optimality of the EW portfolio (i.e. $\sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_{\eta}^{\mathbb{P}}(\mathbf{x}^u) \leq \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_{\eta}^{\mathbb{P}}(\mathbf{x}), \forall \mathbf{x} \in \mathcal{X}$) by some arithmetic manipulations:

$$\sum_{k \in [K]} w_k \theta_k \geq \frac{\|\mathbf{x} - \mathbf{x}^u\|_{m^*}}{\|\mathbf{x}\|_{m^*} - \|\mathbf{x}^u\|_{m^*}} \mathbb{E}^{\hat{\mathbb{P}}}[\|\mathbf{r}\|_m], \forall \mathbf{x} \in \mathcal{X}.$$

Therefore, as $\sum_{k \in [K]} w_k \theta_k \rightarrow \infty$, this sufficient condition would finally hold. Especially like (Pflug et al. 2012, Proposition 3), the solution of our DRO problem would converge to EW portfolio under ℓ_1 -norm if $\sum_{k \in [K]} w_k \theta_k \geq (I - 1) \mathbb{E}^{\hat{\mathbb{P}}}[\|\mathbf{r}\|_1]$. $\square$

EC.2. Proof in Section 4

Before showing the proof, we first state the following several lemmas and definitions.

EC.2.1. Additional Definitions of Markov Chain

We follow the same definition in Assumption 1 and denote some other definitions extracted from Wolfer and Kontorovich (2021). Suppose the underlying true transition matrix of this Markov Chain is A^* .

DEFINITION EC.1 (SPECTRAL GAP). The *reversible* transition matrix A^* satisfies *detailed balance*, i.e. $\pi_i A_{i,j}^* = \pi_j A_{j,i}^*, \forall i, j \in [K]$ with π being the stationary distribution. Then for the eigenvalues of A^* : $1 = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_K$, the spectral gap is defined by:

$$\gamma_s = 1 - \lambda_2(A^*).$$

DEFINITION EC.2 (PSEUDO-SPECTRAL GAP). Pseudo-spectral gap is defined by:

$$\gamma_{ps} := \max_{m \geq 1} \left\{ \frac{\gamma_s((A^+)^m (A^*)^m)}{m} \right\},$$

where A^+ is the time reversal of A , given by $A_{i,j}^+ := \frac{\pi_j A_{j,i}^*}{\pi_i}, \forall i, j \in [K]$.

EC.2.2. Technical Lemmas

LEMMA EC.3 (Estimation Error of Transition Matrix of Markov Chain). Under Assumption 1, with probability $1 - \delta$, as long as the window size N is large enough ($N \geq \frac{C}{\gamma_{ps}\pi_*} \log(\frac{K}{\delta} \sqrt{\frac{1}{\pi_*}})$), the output of the frequency-based approach w_i satisfies:

$$|w_i - w_i^*| \leq \sqrt{\frac{CK \log(K/\delta)}{N\pi_*}}, \forall i \in [K], \quad (\text{EC.9})$$

where C is the same constant in Theorem 3.1 in Wolfer and Kontorovich (2021).

Proof of Lemma EC.3. Recall $\pi_* = \min_{i \in [K]} \pi_i$ in Assumption 1, this result is a variant of Theorem 3.1 in Wolfer and Kontorovich (2021). The empirical estimator used in the proof of Theorem 3.1 in Wolfer and Kontorovich (2021) is exactly the estimation procedure of the probability transition matrix A shown in Section 4.1.1 as well as Step 3(a) in Algorithm 1. Therefore, compared with the underlying true transition matrix $A^* = \{a_{j,k}^*\}_{j,k \in [K]}$ of regime (state) sequences in Assumption 1, since the sample size $N \geq \frac{C}{\gamma_{ps}\pi_*} \log(\frac{K}{\delta} \sqrt{\frac{1}{\pi_*}})$, then whenever $N \geq C \frac{K \log(K/\delta)}{\varepsilon^2 \pi_*}$, we have with probability at least $1 - \delta$, $\|A - A^*\|_\infty \leq \varepsilon$, which implies that $\|A - A^*\|_\infty \leq \sqrt{C} \sqrt{\frac{K \log(K/\delta)}{N\pi_*}}$ under large N with probability at least $1 - \delta$. Finally, by definition of matrix norm, $\|A - A^*\|_\infty = \max_j \sum_{k \in [K]} |a_{j,k} - a_{j,k}^*|$ and $w_j = a_{i^*,j}$ for the last observed regime i^* , we can obtain the bound in (EC.9). $\square$

LEMMA EC.4 (Empirical State Frequency Concentration). Let $\{s_i\}_{i \geq 1}$ be a reversible Markov chain with stationary distribution π and spectral gap $\gamma_s > 0$. Then denote $N_k = \sum_{i=1}^N \mathbb{I}_{\{s_i=k\}}$, we have:

$$\Pr \left(\left| \frac{N_k}{N} - \pi_k \right| \geq \varepsilon \right) \leq 2 \exp(-2c'_0 N \varepsilon^2), \quad (\text{EC.10})$$

where c'_0 is the same as defined in Proposition 3.

Proof of Lemma EC.4. This result is a direct application from Theorem 2.1 in Fan et al. (2021) by letting the bounded function $f_i(s) = \mathbb{I}_{\{s=k\}}, \forall i \in [N]$ in their condition. Then $\mathbb{E}^\pi[f_i(s)] = \pi_k, \forall i \in [N]$; $\sum_{i=1}^N f_i(s_i) = N_k$ so that we first obtain from Theorem 2.1 in Fan et al. (2021), $\forall \varepsilon' > 0$:

$$\Pr(|N_k - N\pi_k| \geq \varepsilon') \leq 2 \exp \left(-\frac{2c'_0 \varepsilon'^2}{N} \right). \quad (\text{EC.11})$$

Letting $\varepsilon = N\varepsilon'$, we would obtain the result of (EC.10). $\square$

LEMMA EC.5 (Some Properties of Wasserstein Distance). *We need the following properties of Wasserstein Distance in our analysis:*

- (a): *Convexity (Lemma 2.10 in Pflug and Pichler (2014)):*

$$W_m(\mathbb{P}, (1-\lambda)\mathbb{P}_0 + \lambda\mathbb{P}_1) \leq (1-\lambda)W_m(\mathbb{P}, \mathbb{P}_0) + \lambda W_m(\mathbb{P}, \mathbb{P}_1), \forall \lambda \in [0, 1].$$

- (b): *Measure Concentration (Theorem 3.4 in Esfahani and Kuhn (2018)):* if $\mathbb{P} \in \mathcal{P}(\mathbb{R}^d)$ ($d \geq 2$) is a light-tailed distribution (i.e. satisfying Assumption 3), ξ_i i.i.d. each sampling from $\mathbb{P}$ and $\hat{\mathbb{P}}_n = \frac{1}{n} \sum_{i \in [n]} \delta_{\xi_i}$, then $\forall t \in [0, 1]$:

$$\Pr(W_m(\mathbb{P}, \hat{\mathbb{P}}_n) \geq t) \leq c_1 \exp(-c_2 n t^d),$$

where c_1, c_2 are positive constants only depending on $\mathbb{P}$.

- (c): *Variational Representation (Theorem 3.2 in Esfahani and Kuhn (2018)):* for any distribution $\mathbb{P}_1, \mathbb{P}_2 \in \mathcal{P}(\mathbb{R}^d)$,

$$W_m(\mathbb{P}_1, \mathbb{P}_2) = \sup_{f \in \mathcal{L}_m} (\mathbb{E}^{\mathbb{P}_1}[f(\xi)] - \mathbb{E}^{\mathbb{P}_2}[f(\xi)]),$$

where $\mathcal{L}_m$ denotes the space of all Lipschitz functions with $|f(\xi_1) - f(\xi_2)| \leq \|\xi_1 - \xi_2\|_m, \forall \xi_1, \xi_2 \in \mathbb{R}^d$.

LEMMA EC.6 (Feasibility of Transportation Problem). *The following linear systems with variables $\{\mu_{ik}\}_{i \in [I], k \in [K]}$ is feasible if $\{w_i^*\}_{i \in [I]}$ and $\{w_k\}_{k \in [K]}$ are all nonnegative and $\sum_{i \in [I]} w_i^* = \sum_{k \in [K]} w_k$:*

$$\begin{aligned} \sum_{k \in [K]} \mu_{ik} &= w_i^*, \quad \forall i \in [I]; \\ \sum_{i \in [I]} \mu_{ik} &= w_k, \quad \forall k \in [K]; \\ \mu_{ik} &\geq 0, \quad \forall i \in [I], k \in [K]. \end{aligned}$$

This lemma follows directly from Chapter 6 in Luenberger et al. (1984) since it can be regarded as the feasible region of one balanced transportation problem.

The following lemma serves as a building block to show Theorem 2, which decomposes $\mathbb{P}^*$ with a mixture of “proper distributions” $\hat{\mathbb{P}}_k^*$ with their regime frequencies by $\{w_k\}_{k \in [K]}$ instead of the true $\{w_k^*\}_{k \in [K]}$.

LEMMA EC.7 (Probability Reconstruction). *We can represent the underlying true distribution $\mathbb{P}^*$ to be $\mathbb{P}^* = \sum_{k \in [K]} w_k \hat{\mathbb{P}}_k^*$, where $\hat{\mathbb{P}}_k^* = \sum_{i \in [K]} \alpha_{ik} \mathbb{P}_i^*, \forall k \in [K]$ with a set of nonnegative $\{\alpha_{ik}\}_{i, k \in [K]}$ and $\sum_{i \in [K]} \alpha_{ik} = 1, \forall k \in [K]$. And $\forall k \in [K]$,*

$$\alpha_{kk} = \begin{cases} 1, & \text{if } w_k \leq w_k^* \\ \frac{w_k^*}{w_k}, & \text{if } w_k > w_k^* \end{cases}. \quad (\text{EC.12})$$

Proof of Lemma EC.7. Since $\mathbb{P}^* = \sum_{k \in [K]} w_k^* \mathbb{P}_k^*$ from Assumption 2, if we want $\mathbb{P}^{**} = \sum_{k \in [K]} w_k \hat{\mathbb{P}}_k^*$, we need to show that we can represent $w_k^* = \sum_{i \in [K]} \alpha_{ki} w_i, \forall k \in [K]$. It is equivalent to say that α_{ik} need to satisfy the following constraints:

$$\begin{aligned} \sum_{k \in [K]} \alpha_{ik} w_k &= w_i^*, \quad \forall i \in [K]; \\ \sum_{i \in [K]} \alpha_{ik} &= 1, \quad \forall k \in [K]; \\ \alpha_{ik} &\geq 0, \quad \forall i, k \in [K]. \end{aligned} \quad (\text{EC.13})$$

where the second and third set of constraints is to guarantee that the new constructed $\hat{\mathbb{P}}_k^* := \sum_{i \in [K]} \alpha_{ik} \mathbb{P}_i^*$ is a distribution. We need to show (EC.13) is feasible for $\{\alpha_{ik}\}_{i,k \in [K]}$ satisfying (EC.12). That is to say, we need to find a set of feasible solution of $\{\alpha_{ik}\}_{i,k \in [K]}$ in (EC.13) satisfying (EC.12).

Denote $\tilde{K} = \{k | w_k = 0, k \in [K]\}$. If $\tilde{K} \neq \emptyset$, for those variables $\{\alpha_{ik}\}_{i \in [K], k \in \tilde{K}}$, we assign $\alpha_{ik} = \begin{cases} 1, & \text{if } i = k \\ 0, & \text{otherwise} \end{cases}$ such that the assigned $\{\alpha_{ik}\}_{i \in [K], k \in \tilde{K}}$ satisfy (EC.12) and feasible in (EC.13) for the constraints they appear. Then we need to find one feasible $\{\alpha_{ik}\}_{i \in [K], k \in [K] \setminus \tilde{K}}$ to satisfy (EC.12) and (EC.13) under the value assignments of $\{\alpha_{ik}\}_{i \in [K], k \in \tilde{K}}$.

Since $w_k > 0$ for $k \in [K] \setminus \tilde{K}$, if we define $\mu_{ik} = w_k \alpha_{ik}, \forall i, k \in [K] \setminus \tilde{K}$, then the feasibility of $\{\alpha_{ik}\}_{i \in [K], k \in [K] \setminus \tilde{K}}$ in (EC.13) is equivalent to that of $\{\mu_{ik}\}_{i \in [K], k \in [K] \setminus \tilde{K}}$ due to one to one correspondence with the following constraints:

$$\begin{aligned} \sum_{k \in [K] \setminus \tilde{K}} \mu_{ik} &= w_i^*, \quad \forall i \in [K]; \\ \sum_{i \in [K]} \mu_{ik} &= w_k, \quad \forall k \in [K] \setminus \tilde{K}; \\ \mu_{ik} &\geq 0, \quad \forall i \in [K], k \in [K] \setminus \tilde{K}. \end{aligned} \tag{EC.14}$$

Based on the discussion above, it is equivalent to showing that there is a set of feasible solution $\{\mu_{ik}\}_{i \in [K], k \in [K] \setminus \tilde{K}}$ where $\mu_{kk} = \min\{w_k, w_k^*\}, \forall k \in [K] \setminus \tilde{K}$ in (EC.14).

Define $I_1 = \{i \in [K] : w_i \geq w_i^*\}$, $K_1 = \{k \in [K] : 0 < w_k < w_k^*\}$. Then $I_1 \cup K_1 = [K] \setminus \tilde{K}$, $I_1 \cap K_1 = \emptyset$. Since $\mu_{kk} = \min\{w_k, w_k^*\}, \forall k \in [K] \setminus \tilde{K}$, by (EC.14) we have:

- If $i \in I_1$, then $\mu_{ii} = w_i^*$; $\mu_{ik} = 0, \forall k \neq i, k \in [K] \setminus \tilde{K}$;
- If $k \in K_1$, then $\mu_{kk} = w_k$; $\mu_{ik} = 0, \forall i \neq k, i \in [K]$.

Then by computation, the remaining unassigned $\{\mu_{ik}\}_{i \in K_1 \cup \tilde{K}, k \in I_1}$ would need to satisfy following constraints under our assignments of other variables above:

$$\begin{aligned} \sum_{k \in I_1} \mu_{ik} &= w_i^* - w_i, \quad \forall i \in K_1 \cup \tilde{K}; \\ \sum_{i \in K_1 \cup \tilde{K}} \mu_{ik} &= w_k - w_k^*, \quad \forall k \in I_1; \\ \mu_{ik} &\geq 0, \quad \forall i \in K_1 \cup \tilde{K}, k \in I_1. \end{aligned} \tag{EC.15}$$

And the region (EC.15) is feasible for $\{\mu_{ik}\}_{i \in K_1 \cup \tilde{K}, k \in I_1}$ since $\sum_{i \in K_1 \cup \tilde{K}} (w_i^* - w_i) = \sum_{k \in I_1} (w_i - w_i^*) > 0$ by definition of $\tilde{K}, I_1, K_1$ from Lemma EC.6. Take one feasible solution $\{\mu_{ik}\}_{i \in K_1 \cup \tilde{K}, k \in I_1}$ and by previous assignments, we find a feasible solution for $\{\mu_{ik}\}_{i \in [K], k \in [K] \setminus \tilde{K}}$ satisfying (EC.14) with $\mu_{kk} = \min\{w_k, w_k^*\}, \forall k \in [K] \setminus \tilde{K}$.

Then for any feasible $\{\mu_{ik}\}_{i \in [K], k \in [K] \setminus \tilde{K}}$ above, it corresponds to one feasible $\{\alpha_{ik}\}_{i \in [K], k \in [K] \setminus \tilde{K}}$ under our assignment for $\{\alpha_{ik}\}_{i \in [K], k \in \tilde{K}}$. Therefore, we obtain a set of feasible solution of $\{\alpha_{ik}\}_{i,k \in [K]}$ in (EC.13) satisfying (EC.12) and show $\mathbb{P}^* = \sum_{k \in [K]} w_k (\sum_{i \in [K]} \alpha_{ik} \mathbb{P}_i^*)$ for some $\{\alpha_{ik}\}_{i,k \in [K]}$ satisfying (EC.12). $\square$

EC.2.3. Proof of Main Results

Proof of Theorem 2. By Lemma EC.7, we can represent $\mathbb{P}^* = \sum_{k \in [K]} w_k \hat{\mathbb{P}}_k^*$ where $\hat{\mathbb{P}}_k^* = \sum_{i \in [K]} \alpha_{ik} \mathbb{P}_i^*, \forall k \in [K]$ with the same requirement for $\{\alpha_{ik}\}_{i,k \in [K]}$.

Then for each regime occurring with probability w_k , by the representation of our ambiguity set in Lemma EC.1, we only need to consider $\theta_k \geq W_m(\mathbb{P}_k, \hat{\mathbb{P}}_k^*)$ to cover the true distribution. And by triangle inequality, we have:

$$W_m(\hat{\mathbb{P}}_k^*, \hat{\mathbb{P}}_k) \leq W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k) + W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k^*) \tag{EC.16}$$

We aim to bound the two terms of right-hand side with high probability.

For the first term of right-hand side in (EC.16), we leverage on the result of the measure concentration of Wasserstein distance (i.e. Lemma EC.5 (b)). Specially, $\hat{\mathbb{P}}_k$ is the empirical distribution from $\mathbb{P}_k^*$ and we have:

$$\Pr(W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k) \geq t) \leq \hat{c}_1 \exp(-\hat{c}_2 N_k t^I), \quad (\text{EC.17})$$

where $\hat{c}_1, \hat{c}_2$ are positive constants only depending on $\mathbb{P}_k^*$. Then equating the right-hand side of (EC.17) to $\frac{\delta}{2K}$ and solving for t yields $t = \left(\frac{\log(2K\hat{c}_1/\delta)}{\hat{c}_2 N_k} \right)^{\frac{1}{I}}$. Then letting $c_{k,1} = \frac{1}{\hat{c}_2}, c_{k,2} = \log(2\hat{c}_1)$, we would have: $W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k) \leq \left(\frac{c_{k,1} \log(K/\delta) + c_{k,2}}{N_k} \right)^{\frac{1}{I}}$ with probability at least $1 - \frac{\delta}{2K}$. Therefore, denote the event $A_k = \left\{ W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k) \leq \left(\frac{c_{k,1} \log(K/\delta) + c_{k,2}}{N_k} \right)^{\frac{1}{I}} \right\}$, we have $\Pr(A_k) \geq 1 - \frac{\delta}{2K}, \forall k \in [K]$.

For the second term of right-hand side in (EC.16), if $w_k = 0$, then $\hat{\mathbb{P}}_k^* = \mathbb{P}_k^*$ by (EC.12) and the second term would disappear. Otherwise for $w_k \neq 0$, we have:

$$W_m(\mathbb{P}_k^*, \hat{\mathbb{P}}_k^*) = W_m(\mathbb{P}_k^*, \sum_{i \in [K]} \alpha_{ik} \mathbb{P}_i^*) \leq \sum_{i \in [K], i \neq k} \alpha_{ik} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*) \leq (1 - \min_{k \in [K]} \min\{\frac{w_k^*}{w_k}, 1\}) \max_{i \in [K]} W_m(\mathbb{P}_i^*, \mathbb{P}_k^*), \quad (\text{EC.18})$$

where the first inequality follows from Lemma EC.5(a). Specially by repeating using convexity property, we have:

$$\begin{aligned} W_m(\mathbb{P}_k^*, \sum_{i \in [K]} \alpha_{ik} \mathbb{P}_i^*) &= W_m(\mathbb{P}_k^*, \alpha_{1k} \mathbb{P}_1^* + (1 - \alpha_{1k}) \sum_{i > 1} \frac{\alpha_{ik} \mathbb{P}_i^*}{1 - \alpha_{1k}}) \\ &\leq \alpha_{1k} W_m(\mathbb{P}_k^*, \mathbb{P}_1^*) + (1 - \alpha_{1k}) W_m(\mathbb{P}_k^*, \sum_{i > 1} \frac{\alpha_{ik} \mathbb{P}_i^*}{1 - \alpha_{1k}}) \\ &\leq \alpha_{1k} W_m(\mathbb{P}_k^*, \mathbb{P}_1^*) + \alpha_{2k} W_m(\mathbb{P}_k^*, \mathbb{P}_2^*) + (1 - \alpha_{1k} - \alpha_{2k}) W_m\left(\mathbb{P}_k^*, \sum_{i > 2} \frac{\alpha_{ik} \mathbb{P}_i^*}{1 - \alpha_{1k} - \alpha_{2k}}\right) \\ &\dots \leq \sum_{i \in [K]} \alpha_{ik} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*) = \sum_{i \in [K], i \neq k} \alpha_{ik} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*). \end{aligned}$$

And the second inequality follows by noticing:

$$\begin{aligned} \sum_{i \in [K], i \neq k} \alpha_{ik} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*) &\leq \left(\sum_{i \in [K], i \neq k} \alpha_{ik} \right) \max_{i \in [K]} W_m(\mathbb{P}_i^*, \mathbb{P}_k^*) \\ &\leq (1 - \min\{\frac{w_k^*}{w_k}, 1\}) \max_{i \in [K]} W_m(\mathbb{P}_i^*, \mathbb{P}_k^*), \end{aligned}$$

where the last inequality follows by observing $\sum_{i \in [K], i \neq k} \alpha_{ik} = 1 - \alpha_{kk} = 1 - \min\{\frac{w_k^*}{w_k}, 1\} \leq 1 - \min_{k \in [K]} \min\{\frac{w_k^*}{w_k}, 1\}$.

Furthermore, we have:

$$\begin{aligned} \min_{k \in [K]} \min\left\{\frac{w_k^*}{w_k}, 1\right\} &= 1 + \min_{k \in [K]} \min\left\{\frac{w_k^* - w_k}{w_k}, 0\right\} \\ &\geq 1 - \max_{k \in [K]} \frac{|w_k^* - w_k|}{w_k} \geq 1 - \frac{\max_{k \in [K]} |w_k - w_k^*|}{\min_{i \in [K], w_i \neq 0} w_i} \end{aligned}$$

From Lemma EC.3, if the sample size $N \geq \frac{c_0^2}{\gamma_{ps} \pi_*} \log\left(\frac{2K}{\delta} \sqrt{\frac{1}{\pi_*}}\right)$, then with probability at least $1 - \frac{\delta}{2}$, $|w_k - w_k^*| \leq c_0 \sqrt{\frac{K \log(2K/\delta)}{N \pi_*}}, \forall k \in [K]$, which implies that: $\min_{k \in [K]} \min\{\frac{w_k^*}{w_k}, 1\} \geq 1 -$

$\frac{c_0}{\min_{i \in [K], w_i \neq 0} w_i} \sqrt{\frac{K \log(2K/\delta)}{N\pi_*}}$. Therefore, denote the event $B = \left\{ |w_k - w_k^*| \leq c_0 \sqrt{\frac{K \log(2K/\delta)}{N\pi_*}}, \forall k \in [K] \right\}$. Then $\Pr(B) \geq 1 - \frac{\delta}{2}$, and under the event B , we have: $\min_{k \in [K]} \min\{\frac{w_k^*}{w_k}, 1\} \geq 1 - \frac{c_0}{\min_{i \in [K], w_i \neq 0} w_i} \sqrt{\frac{K \log(2K/\delta)}{N\pi_*}}$.

Finally, under the event $(\bigcap_{k \in [K]} A_k) \cap B$, following the decomposition in (EC.16) and (EC.18), we have:

$$W_m(\hat{\mathbb{P}}_k^*, \hat{\mathbb{P}}_k) \leq \left(\frac{c_{k,1} \log(K/\delta) + c_{k,2}}{N_k} \right)^{\frac{1}{T}} + \frac{c_0}{\min_{i \in [K], w_i \neq 0} w_i} \sqrt{\frac{K \log(2K/\delta)}{N\pi_*}} \max_{i \neq k} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*), \forall k \in [K]. \quad (\text{EC.19})$$

Therefore, if each component θ_k in $\boldsymbol{\theta}$ is larger than the right-hand side of (EC.19) for each $k \in [K]$, then $\theta_k \geq W_m(\hat{\mathbb{P}}_k^*, \hat{\mathbb{P}}_k)$. And due to $\mathbb{P}^* = \sum_{k \in [K]} w_k \hat{\mathbb{P}}_k^*$ and the representation $\Pi_r \mathcal{U}_{rs}(\boldsymbol{\theta})$ in Lemma EC.1, the true distribution would lie in the ambiguity set, i.e. $\mathbb{P}^* \in \Pi_r \mathcal{U}_{rs}(\boldsymbol{\theta})$.

And we only need to show the event $(\bigcap_{k \in [K]} A_k) \cap B$ happen with probability at least $1 - \delta$ to reach our conclusion. This is given by:

$$\begin{aligned} \Pr\left(\left(\bigcap_{k \in [K]} A_k\right) \cap B\right) &= 1 - \Pr\left(\bigcup_{k \in [K]} A_k^C \cup B^C\right) \\ &\geq 1 - \sum_{k \in [K]} \Pr(A_k^C) - \Pr(B^C) \\ &\geq 1 - K \times \frac{\delta}{2K} - \frac{\delta}{2} = 1 - \delta. \end{aligned}$$

□

Proof of Proposition 3. We follow the proof notation in Theorem 2.

For any $\delta > 0$, under the event $B = \left\{ |w_k - w_k^*| \leq c_0 \sqrt{\frac{K \log(2K/\delta)}{N\pi_*}}, \forall k \in [K] \right\}$, if $N \geq \frac{c_0^2 K \log(2K/\delta)}{\pi_* (\min_{i \in [K]} w_i^*)^2}$, then we have: $|w_k - w_k^*| \leq \min_{i \in [K]} w_i^* \leq w_k^*, \forall k \in [K]$. And $\min_{i \in [K]} w_i \leq 2 \min_{i \in [K]} w_i^*$.

We also denote the event $C_k = \left\{ \left| \frac{N_k}{N} - \pi_k \right| \leq \sqrt{\frac{\log(2K/\delta)}{2c_0' N}} \right\}$ such that the empirical state frequency is close to the stationary distribution. $\Pr(C_K) \geq 1 - \frac{\delta}{K}$ by applying Lemma EC.4 and equating the right-hand of (EC.10) to be $\frac{\delta}{K}$. Then if $\pi_k \geq \sqrt{\frac{\log(2K/\delta)}{2c_0' N}}$, i.e. $N \geq \frac{\log(2K/\delta)}{2c_0' \pi_k^2}$, then $\left| \frac{N_k}{N} - \pi_k \right| \leq \pi_k$ such that $N_k \leq 2N\pi_k$ under event C_k .

Therefore, combining the argument under $(\bigcap_{k \in [K]} A_k) \cap B \cap (\bigcap_{k \in [K]} C_k)$ and the condition of N we have:

$$\begin{aligned} \theta_k &\geq \left(\frac{c_{k,1} \log(K/\delta) + c_{k,2}}{N_k} \right)^{\frac{1}{T}} + \frac{c_0}{\min_{i \in [K], w_i \neq 0} w_i} \sqrt{\frac{K \log(2K/\delta)}{N\pi_*}} \max_{i \neq k} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*) \\ &\geq \left(\frac{c_{k,1} \log(K/\delta) + c_{k,2}}{2N\pi_k} \right)^{\frac{1}{T}} + \frac{c_0}{2 \min_{i \in [K]} w_i^*} \sqrt{\frac{K \log(2K/\delta)}{N\pi_*}} \max_{i \neq k} W_m(\mathbb{P}_k^*, \mathbb{P}_i^*). \end{aligned}$$

And $\Pr((\bigcap_{k \in [K]} A_k) \cap B \cap (\bigcap_{k \in [K]} C_k)) \geq 1 - 2\delta$ by the same argument before. Then replacing δ with $\frac{\delta}{2}$ above would obtain the result in (15). □

Proof of Theorem 3. Denote $\tilde{G}((\mathbf{x}, v), \mathbf{r}) := v + \frac{1}{1-\eta} G(\mathbf{x}, \mathbf{r})$ in (6), then $\text{CVaR}_\eta^\mathbb{P}(\mathbf{x}) = \min_{v \in \mathbb{R}} \mathbb{E}^\mathbb{P}[\tilde{G}(\mathbf{x}, v), \mathbf{r}]$. Suppose $(\mathbf{x}^*, v^*) \in \arg \min_{\mathbf{x}, v} \mathbb{E}^{\mathbb{P}^*}[\tilde{G}((\mathbf{x}, v), \mathbf{r})]$, then $\mathbb{E}^{\mathbb{P}^*}[\tilde{G}((\mathbf{x}^*, v^*), \mathbf{r})] = \text{CVaR}_\eta^{\mathbb{P}^*}(\mathbf{x}^*)$ by definition. Similarly, we define:

$$(\hat{\mathbf{x}}, \hat{v}) \in \arg \min_{\mathbf{x}, v} \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \mathbb{E}^\mathbb{P}[\tilde{G}((\mathbf{x}, v), \mathbf{r})].$$

Notice that $\forall \mathbf{x} \in \mathcal{X}, v \in \mathbb{R}$, we have:

$$|\tilde{G}((\mathbf{x}, v), \mathbf{r}_1) - \tilde{G}((\mathbf{x}, v), \mathbf{r}_2)| \leq \left| \frac{1}{1-\eta} \mathbf{x}'(\mathbf{r}_1 - \mathbf{r}_2) \right| \leq \frac{\|\mathbf{x}\|_{m^*}}{1-\eta} \|\mathbf{r}_1 - \mathbf{r}_2\|_m, \forall m \geq 1.$$

To prove the first inequality, we note that $|\tilde{G}((\mathbf{x}, v), \mathbf{r}_1) - \tilde{G}((\mathbf{x}, v), \mathbf{r}_2)| = \left| \frac{1}{1-\eta} G(\mathbf{x}, \mathbf{r}_1) - \frac{1}{1-\eta} G(\mathbf{x}, \mathbf{r}_2) \right|$. Recall from the definition of $G(\mathbf{x}, \mathbf{r}_1) = (-\mathbf{x}'\mathbf{r}_1 - v)^+$, we can prove the inequality by considering the following three cases:

(i) if $-\mathbf{x}'\mathbf{r}_1 - v \geq 0$ and $-\mathbf{x}'\mathbf{r}_2 - v \leq 0$: $|G(\mathbf{x}, \mathbf{r}_1) - G(\mathbf{x}, \mathbf{r}_2)| = -\mathbf{x}'\mathbf{r}_1 - v \leq -\mathbf{x}'\mathbf{r}_1 + \mathbf{x}'\mathbf{r}_2 \leq |\mathbf{x}'(\mathbf{r}_1 - \mathbf{r}_2)|$.

(ii) if $-\mathbf{x}'\mathbf{r}_1 - v \leq 0$ and $-\mathbf{x}'\mathbf{r}_2 - v \geq 0$: $|G(\mathbf{x}, \mathbf{r}_1) - G(\mathbf{x}, \mathbf{r}_2)| = -\mathbf{x}'\mathbf{r}_2 - v \leq -\mathbf{x}'\mathbf{r}_2 + \mathbf{x}'\mathbf{r}_1 \leq |\mathbf{x}'(\mathbf{r}_1 - \mathbf{r}_2)|$.

(iii) if $-\mathbf{x}'\mathbf{r}_1 - v$ and $-\mathbf{x}'\mathbf{r}_2 - v$ are of the same sign: $|G(\mathbf{x}, \mathbf{r}_1) - G(\mathbf{x}, \mathbf{r}_2)| = |\mathbf{x}'(\mathbf{r}_1 - \mathbf{r}_2)|$.

The second inequality is from the definition of dual norm. Therefore, $\tilde{G}((\mathbf{x}, v), \mathbf{r})$ is Lipschitz continuous with respect to $\mathbf{r}$ with constant upper bounded by $\frac{\|\mathbf{x}\|_{m^*}}{1-\eta}$. Based on Lemma EC.5(c), we have:

$$\mathbb{E}^{\hat{\mathbb{P}}_k^*}[\tilde{G}((\mathbf{x}, v), \mathbf{r})] - \mathbb{E}^{\mathbb{P}_k^*}[\tilde{G}((\mathbf{x}, v), \mathbf{r})] \leq \frac{\|\mathbf{x}\|_{m^*}}{1-\eta} W_m(\hat{\mathbb{P}}_k^*, \mathbb{P}_k^*), \forall k \in [K], \forall \mathbf{x} \in \mathcal{X}, v \in \mathbb{R}.$$

Now we can decompose the generalization error of our DRO model as follows:

$$\begin{aligned} \text{CVaR}_\eta^{\mathbb{P}^*}(\hat{\mathbf{x}}) - \text{CVaR}_\eta^{\mathbb{P}^*}(\mathbf{x}^*) &= (\text{CVaR}_\eta^{\mathbb{P}^*}(\hat{\mathbf{x}}) - \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_\eta^{\mathbb{P}}(\hat{\mathbf{x}})) \\ &\quad + (\sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_\eta^{\mathbb{P}}(\hat{\mathbf{x}}) - \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_\eta^{\mathbb{P}}(\mathbf{x}^*)) + (\sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_\eta^{\mathbb{P}}(\mathbf{x}^*) - \text{CVaR}_\eta^{\mathbb{P}^*}(\mathbf{x}^*)) \\ &\leq \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \text{CVaR}_\eta^{\mathbb{P}}(\mathbf{x}^*) - \text{CVaR}_\eta^{\mathbb{P}^*}(\mathbf{x}^*) \\ &\leq \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \mathbb{E}^{\mathbb{P}}[\tilde{G}((\mathbf{x}^*, v^*), \mathbf{r})] - \mathbb{E}^{\mathbb{P}^*}[\tilde{G}((\mathbf{x}^*, v^*), \mathbf{r})] \\ &= \sum_{k \in [K]} w_k (\sup_{\mathbb{P}_k \in \mathcal{U}_{rs}^k(\theta_k)} \mathbb{E}^{\mathbb{P}_k}[\tilde{G}((\mathbf{x}^*, v^*), \mathbf{r})] - \mathbb{E}^{\hat{\mathbb{P}}_k^*}[\tilde{G}((\mathbf{x}^*, v^*), \mathbf{r})]) \\ &\leq \frac{\|\mathbf{x}^*\|_{m^*}}{1-\eta} \sum_{k \in [K]} w_k \sup_{\mathbb{P}_k \in \mathcal{U}_{rs}^k(\theta_k)} W_m(\mathbb{P}_k, \hat{\mathbb{P}}_k^*) \\ &\leq \frac{\|\mathbf{x}^*\|_{m^*}}{1-\eta} \sum_{k \in [K]} w_k \left(\sup_{\mathbb{P}_k \in \mathcal{U}_{rs}^k(\theta_k)} W_m(\mathbb{P}_k, \hat{\mathbb{P}}_k^*) + W_m(\hat{\mathbb{P}}_k^*, \hat{\mathbb{P}}_k^*) \right) \leq \frac{2\|\mathbf{x}^*\|_{m^*}}{1-\eta} \sum_{k \in [K]} w_k \theta_k. \end{aligned}$$

The first inequality follows from the fact that $\mathbb{P}^* \in \mathcal{U}_{rs}(\boldsymbol{\theta})$ with probability $1 - \delta$ and the optimum of $\hat{\mathbf{x}}$, which reduces the first two terms to non-positive. And the second equality follows from the contained distribution in the ambiguity set $\mathcal{U}_{rs}(\boldsymbol{\theta}) = \{\mathbb{P} : \mathbb{P} = \sum_{k \in [K]} w_k \mathbb{P}_k, \mathbb{P}_k \in \mathcal{U}_{rs}^k(\theta_k), \forall k \in [K]\}$ and $\mathbb{P}^* = \sum_{k \in [K]} w_k \hat{\mathbb{P}}_k^*$. The fourth inequality is from the triangle inequality. The fifth inequality follows from $\hat{\mathbb{P}}_k^* \in \mathcal{U}_{rs}^k(\theta_k)$. $\square$

EC.3. Distributionally Robust Mean-CVaR Model

In the robust Mean-CVaR model, the target is on the *worst-case* expected return of a portfolio; see Kang et al. (2019). In our application, a distributionally robust mean-CVaR portfolio optimization under regime-switching ambiguity set solves the constrained problem (11) with an additional first-moment constraint:

$$\min_{\mathbf{x} \in \mathcal{X}, v \in \mathbb{R}} \left\{ v + \frac{1}{1-\eta} \sup_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \mathbb{E}^{\mathbb{P}}[G(\mathbf{x}, \mathbf{r})] \right\} \quad \text{s.t.} \quad \inf_{\mathbb{P} \in \mathcal{U}_{rs}(\boldsymbol{\theta})} \mathbb{E}^{\mathbb{P}}[\mathbf{r}'\mathbf{x}] \geq R. \quad (\text{EC.20})$$

The following proposition shows that even with worst-case mean constraint, the robust portfolio optimization problem is still tractable. The additional worst-case mean constraint brings an extra set of constraints to the problem (11).

PROPOSITION EC.1. $\forall m \in [1, +\infty]$, the problem (EC.20) can be formulated to be the problem (12) with some additional constraints, given as follows:

$$\begin{aligned} \sum_{k \in [K]} \sum_{n \in [N_k]} \frac{w_k}{N_k} \eta_{nk} - \nu \sum_{k \in [K]} w_k \theta_k &\geq R, \\ \mathbf{r}'_{nk} \mathbf{x} &\geq \eta_{nk}, \forall n \in [N_k], k \in [K], \\ \nu &\geq \|\mathbf{x}\|_{m^*}, \\ \nu &\geq 0, \end{aligned} \tag{EC.21}$$

where $\nu, \eta_{nk} \forall n \in [N_k], k \in [K]$ are additional dual variables introduced.

In particular, when $m = 1$ and $m = 2$, (EC.21) is a set of linear and second-order conic constraints, respectively, which is tractable.

Similarly, if we use the Wasserstein metric with ℓ_2 -norm, the corresponding reformulation is a second-conic program. In our experiments, we will numerically examine both (robust) CVaR and Mean-CVaR portfolios in ℓ_1 -norm.

EC.4. Detailed Cross validation algorithm

Algorithm 4 Specify the ambiguity set size through cross-validation

Input: Return sequences $\mathcal{R}_t$, set of candidate aversion level Θ , number of folds m (assume $\frac{N}{m} \in \mathbb{Z}$), associated observed regime sequences $\mathcal{S}_t := \{s_t, \dots, s_{t+N-1}\}$ (frequency-based) or $\mathcal{O}_t = \{o_t, \dots, o_{t+N-1}\}$ (HMM).

- 1: Partition $\mathcal{R}_t$ into m equally sized subsets with $\mathcal{R}_{t,i} = \{\mathbf{r}_{t+\frac{N(i-1)}{m}}, \dots, \mathbf{r}_{t+\frac{Ni}{m}-1}\}, i \in [m]$.
- 2: **for** $i = 2, \dots, m$ **do**
- 3: **for** $\theta \in \Theta$ **do**
- 4: **for** $j = 0, \dots, \frac{N}{m} - 1$ **do**
- 5: Obtain $\{w_k, \{\mathbf{r}_{nk}\}_{n \in [N_k]}\}_{k \in [K]}$ following the modified Algorithm 2 (input with $\mathcal{D}_t \setminus \mathcal{R}_{t,i}, \mathcal{S}_t \setminus \mathcal{S}_{t,i}$) or Algorithm 3 (input with $\mathcal{R}_t \setminus \mathcal{R}_{t,i}, \mathcal{O}_t \setminus \mathcal{O}_{t,i}$), where the only modification is to set the last regime j^* to $s_{t+\frac{N(i-1)}{m}+j-1}$ in Step 2 of Algorithm 2 or 3.
- 6: Solve the optimization problem in (12) and obtain $\mathbf{x}^*(\theta; \mathcal{R}_t \setminus \mathcal{R}_{t,i}, s_{t+\frac{N(i-1)}{m}+j-1})$.
- 7: **end for**
- 8: **end for**
- 9: Find the best parameter $\theta \in \Theta$ that yields the highest out-of-sample Sharpe ratio (define in Appendix EC.5) on the validation dataset $\mathcal{R}_{t,i}$:

$$\theta_{-i}^* \in \arg \max_{\theta \in \Theta} \text{SharpeRatio} \left(\{\mathbf{r}'_{t+\frac{N(i-1)}{m}+j} \mathbf{x}^*(\theta; \mathcal{R}_t \setminus \mathcal{R}_{t,i}, s_{t+\frac{N(i-1)}{m}+j-1})\}_{j=0}^{N/m-1} \right).$$

10: **end for**

Output: optimal $\theta^* = \frac{1}{m-1} \sum_{i=2}^m \theta_{-i}^*$.

EC.5. Performance Metrics

We set $M = T$ as the time span in the entire dataset and use $N = 120$ and 36 for the DeMiguel et al. (2009)'s datasets and our own dataset, respectively in Algorithm 1. Given the out-of-sample returns, we compute four performance metrics for each portfolio as follows. First, we calculate the out-of-sample Sharpe ratio, defined as the sample mean of the out-of-sample excess returns over the risk-free asset, $\hat{\mu} = \frac{1}{M-N} \sum_{t=1}^{M-N} \hat{g}_{N+t}$, then divided by their sample standard deviation,

$$\hat{\sigma} = \sqrt{\frac{1}{M-N-1} \sum_{t=1}^{M-N} (\hat{g}_{N+t} - \hat{\mu})^2}:$$

$$\text{Sharpe ratio} = \frac{\hat{\mu}}{\hat{\sigma}}.$$

Second, we measure the certainty equivalent (CEQ) return, defined as the risk-free rate that an investor is willing to take compared to using a particular risky strategy. The formula is defined as:

$$\text{CEQ} = \hat{\mu} - \frac{1}{2} \hat{\sigma}^2.$$

Third, we use the maximum drawdown to measure the downside risk of a portfolio over a specific time period. This metric is computed as the maximum observed relative loss from a peak to a trough before attaining a new peak for a given portfolio. Specifically, we use $\{V_k\}_{k=0}^{M-N}$ to denote the accumulated wealth series of the portfolio from the beginning with unit 1, i.e., $V_k = \begin{cases} \prod_{t=1}^k (1 + \hat{g}_{N+t}), & \text{if } k \geq 1, \\ 1, & \text{if } k = 0 \end{cases}$. We define the return peak indices to be $\mathcal{N}_p = \{j \geq 1 | V_j \geq \max\{V_{j-1}, V_{j+1}\}\}$ and denote the element in $\mathcal{N}_p$ as $\{N_p\}_{p \in [|\mathcal{N}_p|+1]}$ with $N_{p+1} = +\infty$. Then, the maximum drawdown is given by:

$$\text{Maximum Drawdown} = 1 - \min_{p \in [|\mathcal{N}_p|]} \frac{\min_{N_p \leq i < \max\{N_{p+1}, M-N\}} V_i}{V_{N_p}}.$$

Fourth, we compute the portfolio turnover to estimate the amount of trading necessary to rebalance each portfolio over time, defined by the average of the ℓ_1 -norm of the trades across the I assets:

$$\text{Turnover} = \frac{1}{M-N-1} \sum_{t=1}^{M-N-1} \|\mathbf{x}_{N+t+1}^* - \mathbf{x}_{N+t+}\|_1,$$

where $\mathbf{x}_{N+t+1}^*$ is the optimal portfolio weight vector at time $N+t+1$ and $\mathbf{x}_{N+t+} \in \mathbb{R}^I$ is the portfolio weight vector right before rebalancing at time $N+t+1$, i.e. for each component $i \in [I]$ of $\mathbf{x}_{N+t+}$, $(\mathbf{x}_{N+t+})_i := \frac{(\mathbf{x}_{N+t})_i(1+(\mathbf{r}_{N+t})_i)}{\sum_{j \in [I]} (\mathbf{x}_{N+t})_j(1+(\mathbf{r}_{N+t})_j)} \in \mathbb{R}$.

EC.6. Additional Empirical Results.

EC.6.1. Empirical Results of DeMiguel et al. (2009)'s Datasets from 1997 to 2019

Since the studies in DeMiguel et al. (2009) are up to 2004, it is interesting to continue the study from 1997 to the first quarter of 2019 such that it covers 2008 financial crisis. We still consider the same assets in those six datasets and conduct a similar study. The only difference is that the data used are recent. Table EC.1 presents the empirical results.

Similarly, the RSDR CVaR (HMM) still uniformly outperforms the EW portfolio in all datasets in terms of Sharpe ratio and CEQ. The maximum drawdowns are comparable among all three portfolios. Again, the EW portfolio contains the least turnover but for some datasets, the turnover rates of the EW and RSDR CVaR (HMM) portfolios are comparable. It is noteworthy that from Table EC.1, RS CVaR (HMM) and DR CVaR (Wasserstein) perform well in some datasets but perform badly in others. RSDR CVaR (HMM) merges their merits to perform generally well in all datasets across different metrics.

The DR and RSDR CVaR portfolios in Tables 2 and EC.1 are obtained using the Wasserstein metric with ℓ_1 -norm. In Appendix EC.6.2, we present another set of results of RSDR CVaR (HMM) using the Wasserstein distance with ℓ_2 -norm. One can observe that the choice of norms does not make a significant difference to the results and our interpretation. Thus, we will stick with ℓ_1 -norm.

In summary, our proposed RSDR CVaR (HMM) beats the EW portfolio in terms of Sharpe ratio and CEQ. It also manifests the key feature of regime-switching model when the regime of the market obviously switches, while enjoying the robustness brought from the DRO framework. To better visualize how the wealth changes over time, we report the time series plots of the portfolio wealth with an initial wealth of unit 1 for the three portfolios in Appendix EC.6.3.

The performance results between regime-switching and non-regime-switching case for CVaR models are similar because all assets in each dataset in DeMiguel et al. (2009) are from the same class. Moreover, assets within one class tend to perform more similarly under the state evolution compared with assets between classes. In the next part, we will show that our HMM models would outperform when the given dataset contains assets from different classes.

Strategy	S&P sectors $I = 11$	Industry sectors $I = 11$	Inter'l portfolios $I = 9$	Mkt/ SMB/HML $I = 3$	FF 1-factor $I = 21$	FF 4-factor $I = 24$
1. Sharpe ratio						
EW	0.1374	0.1802	(0.0873)	0.0730	0.1485	0.1462
RS CVaR (HMM)	0.1637	0.2642	(0.0845)	(0.0644)	0.1689	(0.0409)
DR CVaR (Wasserstein)	0.1413	0.1959	(0.0873)	0.0730	0.1544	0.1555
RSDR CVaR (HMM)	0.1464	0.1989	(0.0873)	0.0730	0.1557	0.1630
2. CEQ						
EW	0.0047	0.0068	(0.0056)	0.0014	0.0064	0.0057
RS CVaR (HMM)	0.0044	0.0085	(0.0043)	(0.0015)	0.0064	(0.0009)
DR CVaR (Wasserstein)	0.0048	0.0069	(0.0056)	0.0014	0.0067	0.0055
RSDR CVaR (HMM)	0.0049	0.0072	(0.0056)	0.0014	0.0067	0.0057
3. Maximum drawdown						
EW	0.4986	0.4848	0.0873	0.2862	0.5380	0.4860
RS CVaR (HMM)	0.4034	0.3455	0.0846	0.2091	0.5249	0.1951
DR CVaR (Wasserstein)	0.5010	0.4438	0.0873	0.2868	0.5214	0.4345
RSDR CVaR (HMM)	0.4929	0.4488	0.0873	0.2859	0.5216	0.4268
4. Turnover						
EW	0.0291	0.0261	0.0253	0.0247	0.0207	0.0246
RS CVaR (HMM)	0.2511	0.3650	0.3446	0.0821	0.3259	0.1902
DR CVaR (Wasserstein)	0.0783	0.0564	0.0253	0.0247	0.0814	0.1781
RSDR CVaR (HMM)	0.1055	0.0894	0.0253	0.0247	0.1794	0.1951

Table EC.1 Empirical Results of DeMiguel et al. (2009)'s datasets from 1997 to 2019 Q1.**EC.6.2. Empirical Results with ℓ_2 -norm in Wasserstein Distance**

For the cost function $\rho(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\|_m$ in (2), we apply ℓ_2 -norm (i.e. $m = 2$) to allow generalizations of our regime-switching DRO model. The results of ℓ_2 -norm for two time periods are shown in Tables EC.2 and EC.3, compared with Tables 2 and EC.1, respectively. Generally, when we consider the same hyperparameter selection for the radius of the ambiguity set θ , we find that the performance of ℓ_2 -norm is similar to that of ℓ_1 -norm but requires much more computation in each oracle.

Metric	S&P sectors $I = 11$	Industry sectors $I = 11$	Inter'l portfolios $I = 9$	Mkt/ SMB/HML $I = 3$	FF 1-factor $I = 21$	FF 4-factor $I = 24$
Sharpe Ratio	0.1838	0.1375	0.1270	0.2259	0.1664	0.1869
CEQ	0.0067	0.0050	0.0046	0.0039	0.0074	0.0072
Maximum drawdown	0.3195	0.4459	0.3907	0.1711	0.4349	0.2729
Turnover	0.0424	0.0287	0.0380	0.0265	0.0268	0.0346

Table EC.2 Empirical Results of DeMiguel et al. (2009)'s datasets during the same time with ℓ_2 -norm.

Metric	S&P sectors $I = 11$	Industry sectors $I = 11$	Inter'l portfolios $I = 9$	Mkt/ SMB/HML $I = 3$	FF 1-factor $I = 21$	FF 4-factor $I = 24$
Sharpe Ratio	0.1449	0.1892	-0.0870	0.0690	0.1515	0.1539
CEQ	0.0049	0.0069	-0.0054	0.0013	0.0065	0.0049
Maximum Drawdown	0.4891	0.4556	0.6193	0.2574	0.5282	0.3661
Turnover	0.0523	0.0470	0.0402	0.0256	0.0603	0.0950

Table EC.3 Empirical Results of DeMiguel et al. (2009)'s datasets from 1997 to 2019 Q1 with ℓ_2 -norm.

EC.6.3. Time Series Plots of the Portfolio Wealth for the Empirical Studies

Figures below show how an initial wealth of unit one varies over time with different portfolios and in different time periods. Figure EC.1 corresponds to the empirical results in Section 5.2 and Figure EC.2 corresponds to the empirical results in Appendix EC.6.1. Note that in some cases, RSDR CVaR (HMM) models is the same as EW portfolio across time and the blue dotted line coincides with the green solid line. We find that our proposed RSDR CVaR (HMM) models can achieve the most portfolio wealth across time periods under most cases.

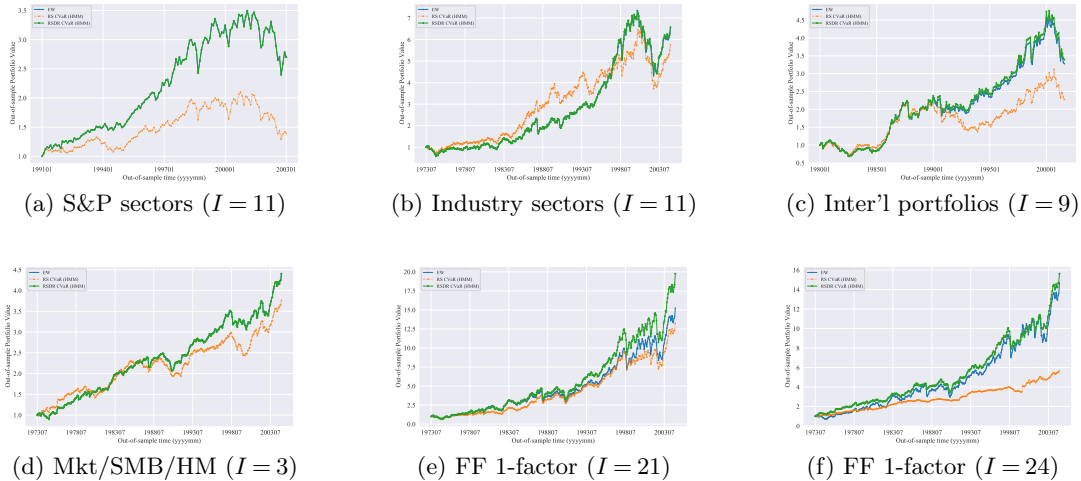


Figure EC.1 Portfolio values of DeMiguel et al. (2009)'s datasets during the same time

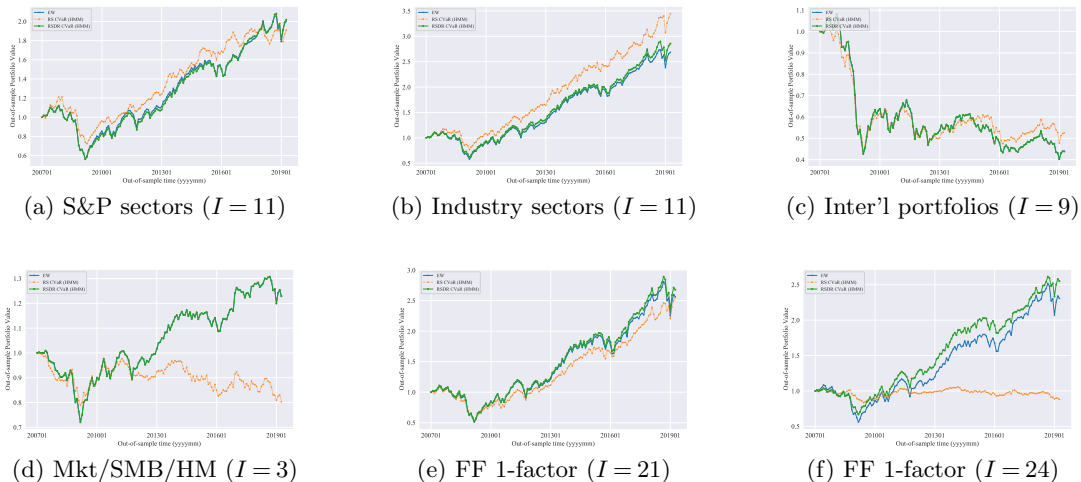
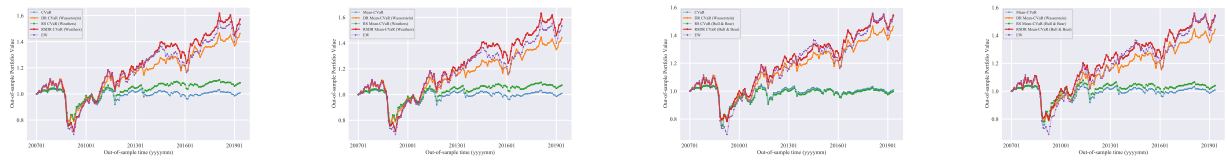


Figure EC.2 Portfolio values of DeMiguel et al. (2009)'s datasets from 1997 to 2019

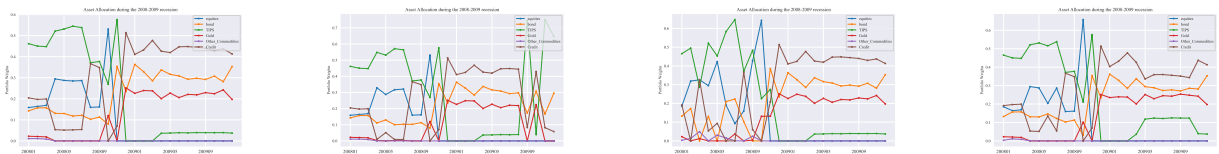
Figures below show how an initial wealth of unit one varies over time corresponding to the empirical results in Section 5.3. Figure EC.3 corresponds to the empirical results of the Weathers and the Bull & Bear Approach.



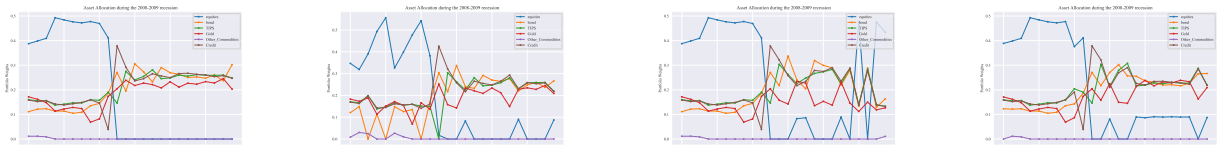
(a) CVaR (Weathers) (b) Mean-CVaR (Weathers) (c) CVaR (Bull & Bear) (d) Mean-CVaR (Bull & Bear)
Figure EC.3 Comparison of different Weathers and Bull & Bear models.

EC.6.4. Asset Allocation for Different Models during the 2008-2009 Great Recession

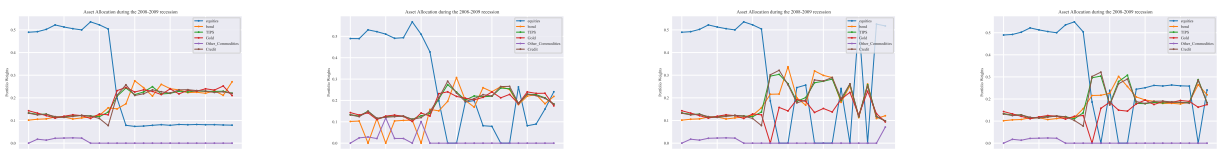
To investigate the effects of parameter sensitivity and regime determination across RSDR CVaR models, we compare different ambiguity set sizes θ for DR-CVaR (Wasserstein), RSDR-CVaR (HMM), RSDR-CVaR (Bull&Bear), and RSDR-CVaR (Weathers) during the 2008-2009 recession in Figures EC.4 ($\gamma = 0$), EC.5 ($\gamma = 0.005$), EC.6 ($\gamma = 0.01$), and EC.7 ($\gamma = 0.025$). In particular, when $\gamma = 0$, the distributionally robust models reduce to the non-robust counterpart.



(a) CVaR (b) RS CVaR (HMM) (c) RS CVaR (Weathers) (d) RS CVaR (Bull&Bear)
Figure EC.4 Asset Allocation for different models during the 2008-2009 great recession when $\gamma = 0$



(a) DR (b) RSDR (HMM) (c) RSDR (Weathers) (d) RSDR (Bull&Bear)
Figure EC.5 Asset Allocation for different models during the 2008-2009 great recession when $\gamma = 0.005$



(a) DR (b) RSDR (HMM) (c) RSDR (Weathers) (d) RSDR (Bull&Bear)
Figure EC.6 Asset Allocation for different models during the 2008-2009 great recession when $\gamma = 0.01$

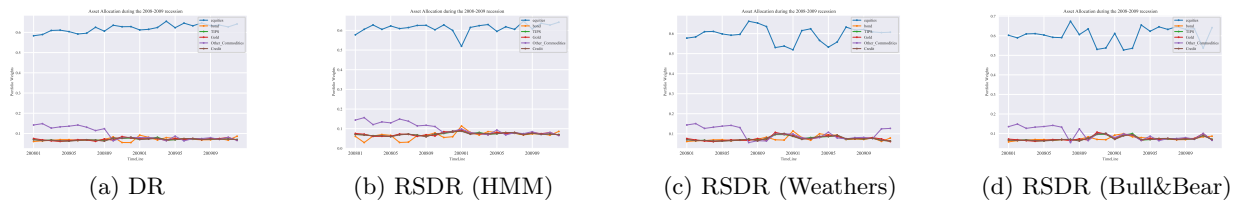


Figure EC.7 Asset Allocation for different models during the 2008-2009 great recession when $\gamma = 0.05$

One can see that when we increase the tuning parameter m to 0.025, the asset allocation would have little variation across time periods (and thus increasing m is somehow practically meaningless). For each fixed m (or equivalently θ), we can observe that the RSDR-CVaR models are generally more volatile than the DR-CVaR model over time, which indicates that regime-switching model considers different mixtures of underlying distributions for asset returns over time and thus can capture the time-varying uncertainties. The comparison between DR and RSDR models is subtle, which can be ascribed to the small window size we chose. When m is not large (i.e., the proposed RSDR models do not converge to EW portfolio), our RSDR models can identify regime-switching behavior based on the negative returns of equities and excess positive returns of bond.